04 Salt Analysis

Introduction:

Qualitative analysis of inorganic salts means the identification of cations and anions present in the salt or a mixture of salts. Inorganic salts may be obtained by complete or partial neutralisation of acid with base or vice-versa. In the formation of a salt, the part contributed by the **acid** is called **anion** (acidic radicals) and the part contributed by the **base** is called **cation** (basic radicals). For example, in the salts $CuSO_4$ and NaCl, $Cu^{2\oplus}$ and $Na^{\oplus}$ ions are respectively cations and $SO_4^{2\ominus}$ and $Cl^{\ominus}$ ions are anions respectively. Qualitative analysis is carried out through the reactions which are easily perceptible to our senses such as sight and smell. Such reactions involve:

(a) Formation of a precipitate

(b) Change in colour

(c) Evolution of gas etc.

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Systematic analysis of an inorganic salt involves the following steps:

(i) Preliminary examination of solid salt and its solution.

(ii) Determination of anions (acidic radicals) by reactions carried out in solution (wet tests) and confirmatory tests.

(iii) Determination of cations (basic radicals) by reactions carried out in solution (wet tests) and confirmatory tests.

Although these tests are not conclusive but sometimes they give quite important clues for the presence of certain anions or cations. These tests can be performed within 10 to 15 minutes. These involve noting the general appearance and physical properties, such as colour, smell, solubility etc. of the salt. These are named as **dry tests.**

Heating of dry salt, blow pipe test, flame tests, borax bead test, sodium carbonate bead test, charcoal cavity test etc. come under dry tests.

Gases evolved in the preliminary tests with dil. H_2SO_4/dil. HCl and conc. H_2SO_4 also give good indication about the presence of acid radicals (See Tables 1 and 3). **Preliminary tests should always be performed before starting the confirmatory tests for the ions.**

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Solubility:

Cation	Anion	Solubility	Exception
Any	$OCl^{\ominus}$, $ClO_2^{\ominus}$, $ClO_3^{\ominus}$, $ClO_4^{\ominus}$, $NO_2^{\ominus}$, $NO_3^{\ominus}$, $CH_3COO^{\ominus}$, $HSO_3^{\ominus}$, $HCO_3^{\ominus}$, $HS^{\ominus}$	All are soluble	(i) $ClO_4^{\ominus}$ of $K^{\oplus}$, $Rb^{\oplus}$, $Cs^{\oplus}$ are insoluble. (ii) $AgNO_2$ is insoluble (iii) CH_3COOAg is partially soluble
$K^{\oplus}$, $Rb^{\oplus}$, $Cs^{\oplus}$ & $NH_4^{\oplus}$	Any	All are soluble	$[Pt\,Cl_6]^{2\ominus}$ & $[Co(NO_2)_6]^{3\ominus}$ are insoluble
$Na^{\oplus}$	Any	All are soluble	----------
Any	$Cl^{\ominus}$, $Br^{\ominus}$, $I^{\ominus}$	All are soluble	(i) $Ag^{\oplus}$, $Pb^{2\oplus}$, $Hg_2^{2\oplus}$, $Cu_2^{2\oplus}$ are insoluble (ii) $PbCl_2$, $CuBr_2$, Cu_2I_2 are soluble on warming (iii) $HgBr_2$ & HgI_2 remain insoluble on warming.
Any	$SO_4^{2\ominus}$	All are soluble	(i)$Ba^{2\oplus}$, $Sr^{2\oplus}$, $Pb^{2\oplus}$ are insoluble (ii) $CaSO_4$, Ag_2SO_4, $SnSO_4$ & $HgSO_4$ are partially soluble
Any	$SO_3^{2\ominus}$, $CO_3^{2\ominus}$, $C_2O_4^{2\ominus}$, $PO_4^{3\ominus}$ $O^{2\ominus}$,$OH^{\ominus}$, $F^{\ominus}$	All are Insoluble	(i) BeF_2 & AgF are soluble (ii) $Na^{\oplus}$ to $Cs^{\oplus}$ & $NH_4^{\oplus}$ are soluble (iii) $O^{2\ominus}$ & $OH^{\ominus}$ of $Ca^{2\oplus}$ $Sr^{2\oplus}$ & $Ba^{2\oplus}$ are soluble
Any	$CN^{\ominus}$, $OCN^{\ominus}$, $SCN^{\ominus}$, $S^{2\ominus}$	All are Insoluble	group-I cation & group-II cation are soluble
Any	$Mn_4^{1\ominus}$ is similar to $ClO_4^{\ominus}$	Same as $ClO_4^{\ominus}$	$NaMnO_4$, $KMnO_4$ are soluble

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1. Sulphides are usually insoluble, leaving those of alkali metals and ammonium sulphide, which are soluble. Sulphides of alkaline earth metals and of ammonium are decomposed in aqueous solution.
2. Compounds of alkali metals and ammonium salts are generally soluble.
3. $CO_3^{2\ominus}$, $OH^{\ominus}$, $CrO_4^{2\ominus}$, $PO_4^{3\ominus}$ & $SO_3^{2\ominus}$ are found generally insoluble, leaving those of alkali metals and ammonium cation, which are soluble.

Some soluble salt :

1. All nitrates ($NO_3^{\ominus}$) are water soluble.
2. All acetates ($CH_3COO^{\ominus}$) salt are water soluble. Except – CH_3COOAg(sparingly soluble) $(CH_3COO)_2Cu$ $(CH_3COO)_2Hg_2$ (sparingly soluble) some basic acetates e.g. those of Fe, Al and Cr are insoluble in water
3. All chloride ($Cl^{\ominus}$) are water soluble. Except – AgCl, $PbCl_2$ (soluble in hot water), Hg_2Cl_2, Cu_2Cl_2. BiOCl and SbOCl (white turbidity)
4. All bromides ($Br^{\ominus}$) are water soluble. Except – AgBr, $PbBr_2$ (soluble in boiled water), Hg_2Br_2, Cu_2Br_2
5. All Iodides ($I^{\ominus}$) are water soluble. Except – AgI, PbI_2 (soluble in boiled water), Hg_2I_2, HgI_2, Cu_2I_2, BiI_3, BiOI (orange turbidity).

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6. All sulphates are water soluble. (some basic sulphates such as those of Hg, Bi, Cr are insoluble) Except–Ag_2SO_4(sparingly soluble), $PbSO_4$, (sparingly soluble), $BaSO_4$, $SrSO_4$, $CaSO_4$(sparingly soluble), Li_2SO_4
7. All nitrites ($NO_2^{\ominus}$)salts are water soluble. Except – $AgNO_2$
8. All existing salts of $HCO_3^{\ominus}$ are water soluble. except. $NaHCO_3$ (sparingly soluble)
9. ($S_2O_3^{2\ominus}$) $\Rightarrow$ Most of the thiosulphates those have been prepared are soluble in water $Ag^{\oplus}$, $Pb^{2\oplus}$ and $Ba^{2\oplus}$ (sparingly soluble) $Hg^{2\oplus}$, $Bi^{3\oplus}$, $Cu^{\oplus}$ = insoluble

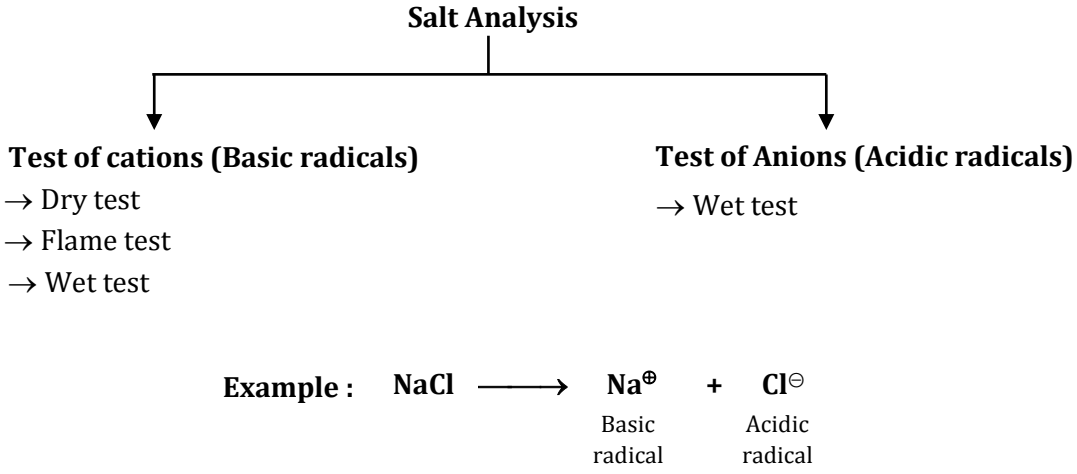
Some Insoluble Salt

1. All carbonates $CO_3^{2\ominus}$ are water insoluble. Except – IA carbonate, Li_2CO_3 (sparingly soluble), $(NH_4)_2CO_3$
2. All sulphites $SO_3^{2\ominus}$ are water insoluble. Except – IA $(NH_4)_2SO_3$
3. All sulphides ($S^{2\ominus}$) are water insoluble.
   Except – IA sulphide, IIA sulphides, $(NH_4)_2S$, (Al, Cr, Mg sulphides are completely hydrolysed)
4. All phosphates ($PO_4^{3\ominus}$) are water insoluble. Except – $(NH_4)_3PO_4$, IA phosphates (except Li_3PO_4)
   IA = 1° 2° 3° soluble IIA = 1° soluble but 2° 3° insoluble.
5. All hydroxides ($OH^{\ominus}$) are water insoluble. Except – IA hydroxide, $Ba(OH)_2$, $Sr(OH)_2$, $Ca(OH)_2$
6. All chromates are water insoluble. Except – Alkali metal $(NH_4)_2CrO_4$, $CaCrO_4$, $MgCrO_4$, $SrCrO_4$(sparingly soluble).
7. All oxalate ($C_2O_4^{2\ominus}$)are water insoluble. Except – IA, Ferrous oxalate, Ammonium oxalate, BeC_2O_4
8. All fluorides ($F^{\ominus}$) are insoluble in water except IA, AgF, HgF_2, AlF_3, BeF_2 and NiF_2, $Pb^{2\oplus}$, $Cu^{2\oplus}$, Fe(III), $Ba^{2\oplus}$, $Li^{\oplus}$ = sparingly soluble

- Their is only hydrated thiosulphate salt is $Na_2S_2O_3 \cdot 5H_2O$.
- All per-magnates ($MnO_4^{\ominus}$)are water soluble.
- $KClO_4$ is not soluble in water.
- $BO_3^{3\ominus}$ $\Rightarrow$ The borates of the alkali metals are readily soluble in water, the borates of the other metals are in general sparingly soluble in water but fairly soluble in acid and in NH_4Cl solution.

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Classification of Radicals:



Example : $NaCl \longrightarrow \underset{\text{Basic radical}}{Na^{\oplus}} + \underset{\text{Acidic radical}}{Cl^{\ominus}}$

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Classification of Anions (Acidic Radicals)

Class – A
(Anions which produce volatile products with H_2SO_4)

Class – B
(Anions which do not produce volatile products with H_2SO_4)

Class – A
(Anions which produce volatile products with H_2SO_4)

Subgroup : I (dil. H_2SO_4) or dil. HCl

$SO_3^{2\ominus}$, $HSO_3^{\ominus}$,
$CO_3^{2\ominus}$, $HCO_3^{\ominus}$,
$CH_3COO^{\ominus}$,
$NO_2^{\ominus}$,
$S_2O_3^{2\ominus}$, $S^{2\ominus}$

Subgroup : II (Conc. H_2SO_4)

$Cl^{\ominus}$, $Br^{\ominus}$, $I^{\ominus}$,
$NO_3^{\ominus}$,
$F^{\ominus}$,
$BO_3^{3\ominus}$,
$C_2O_4^{2\ominus}$

Class – B
(Anions which do not produce volatile products with H_2SO_4)

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Subgroup : I (by ppt)

($SO_4^{2\ominus}$, $PO_4^{3\ominus}$, $CrO_4^{2\ominus}$)

Subgroup : II (by redox)

($CrO_4^{2\ominus}$, $Cr_2O_7^{2\ominus}$, $MnO_4^{\ominus}$)

Class A: Subgroup - 1

Table 1 : Preliminary test with dilute sulphuric acid/dilute hydrochloric acid

Observations	Inference	
	Gas Evolved	Possible Anion
A colourless, odourless gas is evolved with brisk effervescence.	CO_2	Carbonate ($CO_3^{2\ominus}$)
Colourless gas with the smell of rotten eggs is evolved.	H_2S	Sulphide ($S^{2\ominus}$)
Colourless gas with a pungent smell, like burning sulphur.	SO_2	Sulphite ($SO_3^{2\ominus}$)
Brown fumes.	NO_2	Nitrite ($NO_2^{\ominus}$)
Colourless vapours with smell of vinegar.	CH_3COOH vapours	Acetate, ($CH_3COO^{\ominus}$)

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Confirmatory tests for $CO_3^{2\ominus}, S^{2\ominus}, SO_3^{2\ominus}, NO_3^{\ominus}, CH_3COO^{\ominus}$

Confirmatory (wet) tests for anions are performed by using water extract when salt is soluble in water and by using sodium carbonate extract when salt is insoluble in water. Confirmation of $CO_3^{2\ominus}$ is done by using aqueous solution of the salt or by using solid salt as such because sodium carbonate extract contains carbonate ions. Water extract is made by dissolving salt in water. Preparation of sodium carbonate extract is given below.

Preparation of sodium carbonate extract

Take 1 g of salt in a porcelain dish or boiling tube. Mix about 3 g of solid sodium carbonate and add 15 mL of distilled water to it. Stir and boil the content for about 10 minutes. Cool, filter and collect the filtrate in a test tube and label it as sodium carbonate extract.

Confirmatory tests for acid radicals, which react with dilute sulphuric acid are given in Table 2.

Table 2 : Confirmatory tests for $CO_3^{2\ominus}, S^{2\ominus}, SO_3^{2\ominus}, NO_3^{\ominus}, CH_3COO^{\ominus}$

Anion	Confirmatory Test
Carbonate $\left(CO_3^{2\ominus}\right)$	Take 0.1 g of salt in a test tube, add dilute sulphuric acid. CO_2 gas is evolved with brisk effervescence which turns lime water milky. On passing the gas for some more time, milkiness disappears.
Sulphide ($S^{2\ominus}$)	(a) H_2S as evolved turns lead acetate paper black. (b)Take 1 mL of water extract and make it alkaline by adding ammonium hydroxide or sodium carbonate extract. Add a drop of sodium nitroprusside solution. Purple or violet colouration appears.
Sulphite $\left(SO_3^{2\ominus}\right)$	(a) Take 1 mL of water extract or sodium carbonate extract in a test tube and add barium chloride solution. A white precipitate is formed which dissolves in dilute hydrochloric acid and sulphur dioxide gas is also evolved. (b) Take the precipitate of step (a) in a test tube and add a few drops of potassium permanganate solution acidified with dil. H_2SO_4. Colour of potassium permanganate solution gets discharged.
Nitrite $\left(NO_2^{\ominus}\right)$	(a) Take 1 mL of water extract in a test tube. Add a few drops of potassium iodide solution and a few drops of starch solution, acidify with acetic acid. Blue colour appears. (b) Acidify 1 mL of water extract with acetic acid. Add 2-3 drops of sulphanilic acid solution followed by 2-3 drops of 1-naphthylamine reagent. Appearance of red colour indicates the presence of nitrite ion.
Acetate, ($CH_3COO^{\ominus}$)	(a) Take 0.1 g of salt in a china dish. Add 1 mL of ethanol and 0.2 mL conc.H_2SO_4 and heat. Fruity odour confirms the presence of acetate ion. (b) Take 0.1 g of salt in a test tube, add 1-2 mL distilled water, shake well filter if necessary. Add 1 to 2 mL neutralferric chloride solution to the filtrate. Deep red colour appears which disappears on boiling and a brown-red precipitate is formed.

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Chemistry of Confirmatory Tests:

1. Test for Carbonate ion [$CO_3^{2\ominus}$]

If there is effervescence with the evolution of a colourless and odourless gas on adding dil. H_2SO_4 to the solid salt, this indicates the presence of carbonate ion.

The gas turns lime water milky due to the formation of $CaCO_3$

$$Na_2CO_3 + H_2SO_4 \longrightarrow Na_2SO_4 + H_2O + CO_2$$

$$Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$$

If CO_2 gas is passed in excess through lime water, the milkiness disappears due to the formation of calcium hydrogen carbonate which is soluble in water.

$$CaCO_3 + CO_2 + H_2O \longrightarrow Ca(HCO_3)_2$$

2. Test for Sulphide ion [$S^{2\ominus}$]

(a) With warm dilute H_2SO_4 a sulphide gives hydrogen sulphide gas which smells like rotten eggs. A piece of filter paper dipped in lead acetate solution turns black on exposure to the gas due to the formation of lead sulphide which is black in colour.

$$Na_2S + H_2SO_4 \longrightarrow Na_2SO_4 + H_2S$$

$$(CH_3COO)_2Pb + H_2S \longrightarrow \underset{\text{Lead sulphide Black precipitate}}{PbS} + 2CH_3COOH$$

(b) If the salt is soluble in water, take the solution of salt in water make it alkaline with ammonium hydroxide and add sodium nitroprusside solution. If it is insoluble in water take sodium carbonate extract and add a few drops of sodium nitroprusside solution. Purple or violet colouration due to the formation of complex compound $Na_4[Fe(CN)_5NOS]$ confirms the presence of sulphide ion in the salt.

$$Na_2S + \underset{\text{Sodium nitroprusside}}{Na_2[Fe(CN)_5NO]} \longrightarrow \underset{\text{Complex of Purple colour}}{Na_4[Fe(CN)_5NOS]}$$

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3. Test for Sulphite ion [$SO_3^{2\ominus}$]

(a) On treating sulphite with warm dil. H_2SO_4, SO_2 gas is evolved which is suffocating with the smell of burning sulphur.

$$Na_2SO_3 + H_2SO_4 \longrightarrow Na_2SO_4 + H_2O + SO_2$$

The gas turns potassium dichromate paper acidified with dil. H_2SO_4, green.

$$K_2Cr_2O_7 + H_2SO_4 + 3SO_2 \longrightarrow K_2SO_4 + \underset{\substack{\text{Chromium}\\\text{sulphate (green)}}}{Cr_2(SO_4)_3} + H_2O$$

(b) An aqueous solution or sodium carbonate extract of the salt produces a white precipitate of barium sulphite on addition of barium chloride solution.

$$Na_2SO_3 + BaCl_2 \longrightarrow 2NaCl + \underset{\text{White ppt.}}{BaSO_3}$$

This precipitate gives following tests.

(i) This precipitate on treatment with dilute HCl, dissolves due to decomposition of sulphite by dilute HCl. Evolved SO_2 gas can be tested.

$$BaSO_3 + 2HCl \longrightarrow BaCl_2 + H_2O + SO_2$$

(ii) Precipitate of sulphite decolourises acidified potassium permanganate solution.

$$BaSO_3 + H_2SO_4 \longrightarrow BaSO_4 + H_2O + SO_2$$

$$2KMnO_4 + 3H_2SO_4 \longrightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5[O]$$

$$SO_2 + H_2O + [O] \longrightarrow H_2SO_4$$

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4. Test for Nitrite ion [$NO_2^{\ominus}$]

(a) On treating a solid nitrite with dil. H_2SO_4 and warming, reddish brown fumes of NO_2 gas are evolved.

(i) $2NaNO_2 + H_2SO_4 \longrightarrow Na_2SO_4 + 2HNO_2$

$3HNO_2 \longrightarrow HNO_3 + 2NO + H_2O$ (disproportionation)

$2NO + O_2 \longrightarrow 2NO_2$
Brown gas

Addition of potassium iodide solution to the salt solution followed by freshly prepared starch solution and acidification with acetic acid produces blue colour. Alternatively, a filter paper moistened with potassium iodide and starch solution and a few drops of acetic acid turns blue on exposure to the gas, due to the interaction of liberated iodine with starch.

(ii) $NO_2^{\ominus} + CH_3COOH \longrightarrow HNO_2 + CH_3COO^{\ominus}$

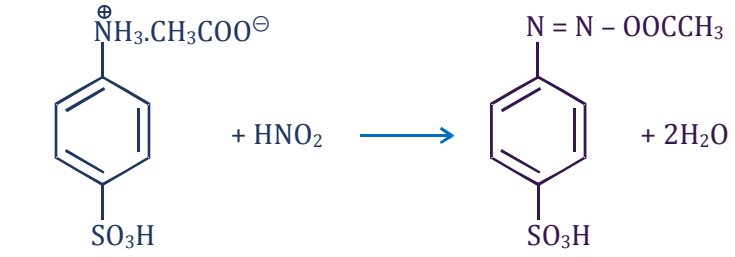
$2HNO_2 + 2KI + 2CH_3COOH \longrightarrow 2CH_3COOK + 2H_2O + 2NO + I_2$

$I_2 + \text{Starch} \longrightarrow$ Blue complex

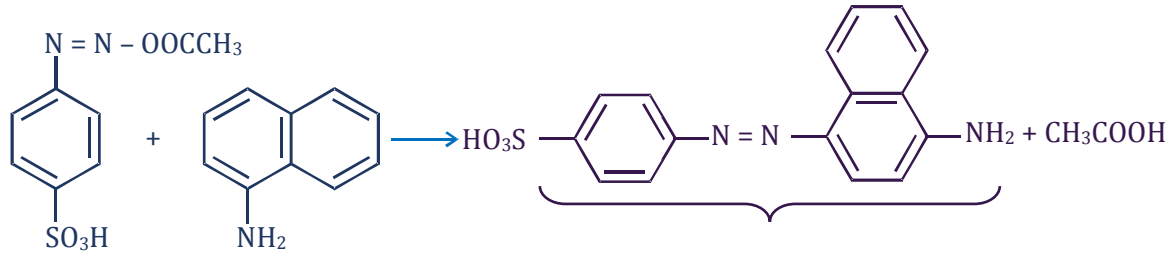
(b) Sulphanilic acid — 1-naphthylamine reagent test (Griess-llosvay test) : On adding sulphanilic acid and 1-naphthylamine reagent to the water extract or acidified with acetic acid, sulphanilic acid is diazotised in the reaction by nitrous acid formed. Diazotised acid couples with 1-naphthylamine to form a red azo-dye.

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$NO_2^{\ominus} + CH_3COOH \rightarrow HNO_2 + CH_3COO^{\ominus}$



(Sulphanilic acid solution)



Diazotised acid | 1-Naphthyl amine | Red azo-dye

The test solution should be very dilute. In concentrated solutions reaction does not proceed beyond diazotisation.

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5. Test for Acetate ion [$CH_3COO^{\ominus}$]

(a) If the salt smells like vinegar on treatment with dil. H_2SO_4, this indicates the presence of acetate ions. Take 0.1 g of salt in a china dish and add 1 mL of ethanol. Then add about 0.2 mL of conc. H_2SO_4 and heat. Fruity odour of ethyl acetate indicates the presence of $CH_3COO^{\ominus}$ ion.

$$2CH_3COONa + H_2SO_4 \longrightarrow Na_2SO_4 + 2CH_3COOH$$

$$CH_3COOH + C_2H_5OH \longrightarrow \underset{\substack{\text{Ethylacetate}\\\text{(Fruity odour)}}}{CH_3COOC_2H_5} + H_2O$$

(b) Acetate gives deep red colour on reaction with neutral ferric chloride solution due to the formation of complex ion which decomposes on heating to give Iron (III) dihydroxyacetate as brown red precipitate.

$$6CH_3COO^{\ominus} + 3Fe^{3\oplus} + 2H_2O \longrightarrow [Fe_3(OH)_2(CH_3COO)_6]^{\oplus} + 2H^{\oplus}$$

$$[Fe_3(OH)_2(CH_3COO^{\ominus})_6]^{\oplus} + 4H_2O \longrightarrow \underset{\substack{\text{Iron(III)dihydroxyacetate}\\\text{(Brown-red precipitate)}}}{3[Fe(OH)_2(CH_3COO)_5]} + 3CH_3COOH + H^{\oplus}$$

6. Test for Acetate ion [$S_2O_3^{2\ominus}$]

(a) Solubility

- Thiosulphates of IA & IIA soluble except BaS_2O_3
- Thiosulphates of $Ag^{\oplus}$, $Pb^{2\oplus}$, $Ba^{2\oplus}$ are sparingly soluble
- Thiosulphates of $Hg^{2\oplus}$, $Bi^{3\oplus}$, Cu^{+} are insoluble

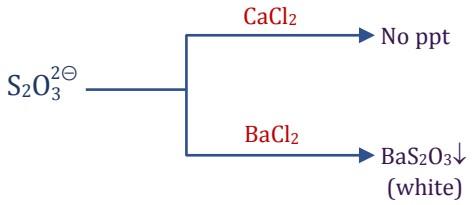
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(b) Test by dil. H_2SO_4 / dil. HCl

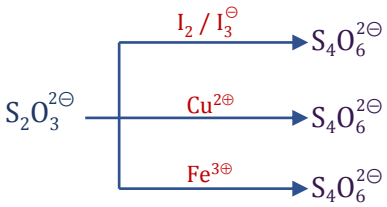
- $S_2O_3^{2\ominus} + 2H^{\oplus} \longrightarrow H_2S_2O_3 \longrightarrow H_2O + S\downarrow + SO_2\uparrow$

(Turbidity)

(c) Test by $CaCl_2$ and $BaCl_2$



(d) Test of $S_2O_3^{2\ominus}$ in presence of weak oxidising agent



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Class A: Subgroup - 2

Table 3 : Preliminary examination with concentrated sulphuric acid

Observations	Inference	
	Gas/Vapours Evolved	Possible Anion
A colourless gas with pungent smell, which gives dense white fumes when a rod dipped in ammonium hydroxide is brought near the mouth of the test tube.	HCl	Chloride, ($Cl^{\ominus}$)
Reddish brown gas with a pungent odour is evolved. Intensity of reddish gas increases on heating the reaction mixture after addition of solid MnO_2 to the reaction mixture. Solution also acquires red colour.	Br_2 vapours	Bromide, ($Br^{\ominus}$)
Violet vapours, which turn starch paper blue and a layer of violet sublimate is formed on the sides of the tube. Fumes become dense on adding MnO_2 to the reaction mixture.	I_2 vapours	Iodide, ($I^{\ominus}$)
Brown fumes evolve which become dense upon heating the reaction mixture after addition of copper turnings and the solution acquires blue colour.	NO_2	Nitrate, ($NO_3^{\ominus}$)
Colourless, odourless gas is evolved which turns lime water milky and the gas coming out of lime water burns with a blue flame, if ignited.	CO and CO_2	Oxalate, $\left(C_2O_4^{2\ominus}\right)$

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Table 4 : Confirmatory tests for $Cl^{\ominus}$, $Br^{\ominus}$, $I^{\ominus}$, $NO_3^{\ominus}$ and $C_2O_4^{2\ominus}$

Anion	Confirmatory Test
Chloride ($Cl^{\ominus}$)	(a) Take 0.1 g of salt in a test tube, add a pinch of manganese dioxide and 3-4 drops of conc. sulphuric acid. Heat the reaction mixture. Greenish yellow chlorine gas is evolved which is detected by its pungent odour and bleaching action. (b) Take 1 mL of sodium carbonate extract in a test tube, acidify it with dil. HNO_3 or take water extract and add silver nitrate solution. A curdy white precipitate is obtained which is soluble in ammonium hydroxide solution. (c) Take 0.1 g salt and a pinch of solid potassium dichromate in a test tube, add conc. H_2SO_4, heat and pass the gas evolved through sodium hydroxide solution. It becomes yellow. Divide the solution into two parts. Acidify one part with acetic acid and add lead acetate solution. A yellow precipitate is formed. Acidify the second part with dilute sulphuric acid and add 1 mL of amyl alcohol followed by 1 mL of 10% hydrogen peroxide. After gentle shaking the organic layer turns blue.
Bromide ($Br^{\ominus}$)	(a) Take 0.1 g of salt and a pinch of MnO_2 in a test tube. Add 3-4 drops conc. sulphuric acid and heat. Intense brown fumes are evolved. (b) Neutralise 1 mL of sodium carbonate extract with hydrochloric acid (or take the water extract). Add 1 mL carbon tetrachloride (CCl_4)/ chloroform ($CHCl_3$)/carbon disulphide. Now add an excess of chlorine water dropwise and shake the test tube. A brown colouration in the organic layer confirms the presence of bromide ion. (c) Acidify 1 mL of sodium carbonate extract with dil. HNO_3 (or take 1 mL water extract) and add silver nitrate solution. A pale yellow precipitate soluble with difficulty in ammonium hydroxide solution is obtained.

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Iodide ($I^{\ominus}$)	(a) Take 1 mL of salt solution neutralised with HCl and add 1 mL chloroform/ carbon tetrachloride/carbon disulphide. Now add an excess of chlorine water drop wise and shake the test tube. A violet colour appears in the organic layer. (b) Take 1 mL of sodium carbonate extract acidify it with dil. HNO_3 (or take water extract). Add, silver nitrate solution. A yellow precipitate insoluble in NH_4OH solution is obtained.

Anion	Confirmatory Test
*Nitrate ($NO_3^{\ominus}$)	Take 1 mL of salt solution in water in a test tube. Add 2 mL of conc. H_2SO_4 and mix thoroughly. Cool the mixture under the tap. Add freshly prepared ferrous sulphate along the sides of the test tube without shaking. A dark brown ring is formed at the junction of the two solutions.
Oxalate ($C_2O_4^{2\ominus}$)	(a) Take 1 mL of water extract or sodium carbonate extract acidified with acetic acid and add calcium chloride solution. A white precipitate insoluble in ammonium oxalate and oxalic acid solution but soluble in dilute hydrochloric acid and dilute nitric acid is formed. (b) Take the precipitate from test (a) and dissolve it in dilute H_2SO_4. Add very dilute solution of $KMnO_4$ and warm. Colour of $KMnO_4$ solution is discharged. Pass the gas coming out through lime water. The lime water turns milky.

Chemistry of Confirmatory Tests:

1. Test for Chloride ion [$Cl^{\ominus}$]

(a) If on treatment with warm conc. H_2SO_4 the salt gives a colourless gas with pungent smell or if the gas which gives dense white fumes with ammonia solution, then the salt may contain $Cl^{\ominus}$ ions and the following reaction occurs.

$$NaCl + H_2SO_4 \longrightarrow \underset{\text{Sodium hydrogen sulphate}}{NaHSO_4} + \underset{\text{Hydrogen chloride Colourless gas}}{HCl}$$

$$HCl + NH_3 \longrightarrow \underset{\text{Ammonium chloride White fumes}}{NH_4Cl}$$

(b) If a salt gives effervescence on heating with conc. H_2SO_4 and MnO_2 and a light greenish yellow pungent gas is evolved, this indicates the presence of $Cl^{\ominus}$ions.

$$MnO_2 + 2NaCl + 2H_2SO_4 \longrightarrow Na_2SO_4 + MnSO_4 + 2H_2O + Cl_2$$

(c) Salt solution acidified with dilute HNO_3 on addition of silver nitrate solution gives a curdy white precipitate soluble in ammonium hydroxide solution. This indicates the presence of $Cl^{\ominus}$ ions in the salt.

$$NaCl + AgNO_3 \longrightarrow NaNO_3 + \underset{\text{Silver chloride (Curdy White precipitate)}}{AgCl}$$

$$AgCl + 2NH_4OH \longrightarrow \underset{\text{Diamine silver (I) chloride (soluble)}}{[Ag(NH_3)_2]Cl} + 2H_2O$$

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(d) Mix a little amount of salt and an equal amount of solid potassium dichromate ($K_2Cr_2O_7$) in a test tube and add conc. H_2SO_4 to it. Heat the test tube and pass the evolved gas through sodium hydroxide solution. If a yellow solution is obtained, divide the solution into two parts. Acidify the first part with acetic acid and then add lead acetate solution. Formation of a yellow precipitate of lead chromate confirms the presence of chloride ions in the salt. This test is called **chromyl chloride test.**

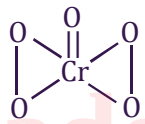
$$4NaCl + K_2Cr_2O_7 + 6H_2SO_4 \longrightarrow 2KHSO_4 + \underset{\text{(Chromyl chloride)}}{2CrO_2Cl_2} + 4NaHSO_4 + 3H_2O$$

$$CrO_2Cl_2 + 4NaOH \longrightarrow Na_2CrO_4 + 2NaCl + 2H_2O$$

$$(CH_3COO)_2Pb + \underset{\text{Sodium chromate}}{Na_2CrO_4} \longrightarrow \underset{\substack{\text{Lead chromate}\\\text{(Yellow precipitate)}}}{PbCrO_4} + 2CH_3COONa$$

Acidify the second part with dilute sulphuric acid and add small amounts of amyl alcohol and then 1 mL of 10% hydrogen peroxide solution. On gentle shaking organic layer turns blue. $CrO_4^{2\ominus}$ ion formed in the reaction of chromyl chloride with sodium hydroxide reacts with hydrogen peroxide to form chromium pentoxide (CrO_5) (See structure) which dissolves in amyl alcohol to give blue colour.

$$CrO_4^{2\ominus} + 2H^{\oplus} + 2H_2O_2 \longrightarrow \underset{\substack{\text{Chromium}\\\text{pentoxide}}}{CrO_5} + 3H_2O$$



Structure of chromium pentoxide

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2. Test for Bromide ion ($Br^{\ominus}$)

If on heating the salt with conc. H_2SO_4 reddish brown fumes of bromine are evolved in excess, this indicates the presence of Br^- ions. The fumes get intensified on addition of MnO_2. Bromine vapours turn starch paper yellow.

$$2NaBr + 2H_2SO_4 \longrightarrow Br_2 + SO_2 + Na_2SO_4 + 2H_2O$$

$$2NaBr + MnO_2 + 2H_2SO_4 \longrightarrow Na_2SO_4 + MnSO_4 + 2H_2O + Br_2$$

(a) Add 1 mL of carbon tetrachloride (CCl_4)/chloroform ($CHCl_3$) and excess of freshly prepared chlorine water dropwise to the salt solution in water or sodium carbonate extract neutralised with dilute HCl. Shake the test tube vigorously. The appearance of an orange brown colouration in the organic layer due to the dissolution of bromine in it, confirms the presence of bromide ions.

$$2NaBr + Cl_2 \longrightarrow 2NaCl + Br_2$$

(b) Acidify the sodium carbonate extract of the salt with dil. HNO_3. Add silver nitrate ($AgNO_3$) solution and shake the test tube. A paleyellow precipitate is obtained which dissolves in ammonium hydroxide with difficulty.

$$NaBr + AgNO_3 \longrightarrow NaNO_3 + \underset{\substack{\text{Silver bromide}\\\text{Pale yellow precipitate}}}{AgBr}$$

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3. Test for Iodide ion ($I^{\ominus}$)

(a) If on heating the salt with conc. H_2SO_4, deep violet vapours with a pungent smell are evolved. These turns starch paper blue and a violet sublimate is formed on the sides of the test tube, it indicates the presence of $I^{\ominus}$ ions. Some HI, sulphur dioxide, hydrogen sulphide, and sulphur are also formed due to the following reactions.

$$2NaI + 2H_2SO_4 \longrightarrow Na_2SO_4 + SO_2 + 2H_2O + I_2$$

$$I_2 + \underset{\text{solution}}{\text{Starch}} \longrightarrow \text{Blue colour}$$

$$NaI + H_2SO_4 \longrightarrow NaHSO_4 + HI$$

$$2HI + H_2SO_4 \longrightarrow 2H_2O + I_2 + SO_2$$

$$6NaI + 4H_2SO_4 \longrightarrow 3I_2 + 4H_2O + S + 3Na_2SO_4$$

$$8NaI + 5H_2SO_4 \longrightarrow 4I_2 + H_2S + 4Na_2SO_4 + 4H_2O$$

On adding MnO_2 to the reaction mixture, the violet vapours become dense.

$$2NaI + MnO_2 + 2H_2SO_4 \longrightarrow I_2 + MnSO_4 + Na_2SO_4 + 2H_2O$$

(b) Add 1 mL of $CHCl_3$ or CCl_4 and chlorine water in excess to the salt solution in water or sodium carbonate extract neutralised with dil. HCl and shake the test tube vigorously. Presence of violet colouration in the organic layer confirms the presence of iodide ions.

$$2NaI + Cl_2 \longrightarrow 2NaCl + I_2$$

Iodine dissolves in the organic solvent and the solution becomes violet.

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(c) Acidify sodium carbonate extract of the salt with dil. HNO_3 and add $AgNO_3$ solution. Appearance of a yellow precipitate insoluble in excess of NH_4OH confirms the presence of iodide ions.

$$NaI + AgNO_3 \longrightarrow \underset{\substack{\text{silver iodide}\\ \text{(Yellow precipitate)}}}{AgI} + NaNO_3$$

4. Test for Fluride ion ($F^{\ominus}$)

(a) Test by conc. H_2SO_4

$$F^{\ominus} + H_2SO_4 \xrightarrow{\Delta} HF\uparrow + HSO_4^{\ominus}$$

(conc.) or

$H_2F_2\uparrow$

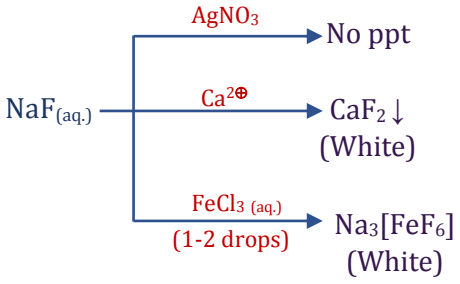
(b) Etching of glass

$$2SiO_2 + 4HF \longrightarrow SiF_4 + 2H_2O$$

$$2SiF_4 + 4H_2O \longrightarrow \underset{\text{(Sillicic acid) Waxy solid}}{H_4SiO_4} + 2HF$$

$$2SiF_4 + 4HF \longrightarrow 2H_2[SiF_6]$$

(c) Precipitation test



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5. Test for nitrate ion ($NO_3^{\ominus}$)

(a) If on heating the salt with conc. H_2SO_4 light brown fumes are evolved then heat a small quantity of the given salt with few copper turnings or chips and conc. H_2SO_4. Evolution of excess of brown fumes indicates the presence of nitrate ions. The solution turns blue due to the formation of copper sulphate.

$$NaNO_3 + H_2SO_4 \longrightarrow NaHSO_4 + HNO_3$$

$$4HNO_3 \longrightarrow 4NO_2 + O_2 + 2H_2O$$

$$2NaNO_3 + 4H_2SO_4 + 3Cu \longrightarrow \underset{\text{Copper sulphate (Blue)}}{3CuSO_4} + Na_2SO_4 + 4H_2O + 2NO$$

$$2NO + O_2 \longrightarrow \underset{\text{(Brown fumes)}}{2NO_2}$$

(b) Take 1 mL of an aqueous solution of the salt and add 2 mL conc. H_2SO_4 slowly. Mix the solutions thoroughly and cool the test tube under the tap. Now, add freshly prepared ferrous sulphate solution along the sides of the test tube dropwise so that it forms a layer on the top of the liquid already present in the test tube. A dark brown ring is formed at the junction of the two solutions due to the formation of nitroso ferrous sulphate (Fig.). Alternatively, first ferrous sulphate is added and then concentrated sulphuric acid is added.



Fig. : Formation of brown ring

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$$NaNO_3 + H_2SO_4 \longrightarrow NaHSO_4 + HNO_3$$

$$6FeSO_4 + 3H_2SO_4 + 2HNO_3 \longrightarrow 3Fe_2(SO_4)_3 + 4H_2O + 2NO$$

$$FeSO_4 + NO \longrightarrow \underset{\text{Nitroso ferrous sulphate (Brown)}}{[Fe(NO)]SO_4}$$

6. Test for Oxalate ion [$C_2O_4^{2\ominus}$]:

If carbon dioxide gas along with carbon monoxide gas is evolved in the preliminary examination with concentrated sulphuric acid, this gives indication about the presence of oxalate ion.

$$Na_2C_2O_4 + Conc.H_2SO_4 \longrightarrow Na_2SO_4 + H_2O + CO_2\uparrow + CO\uparrow$$

Oxalate is confirmed by the following tests:

(a) Acidify sodium carbonate extract with acetic acid and add calcium chloride solution. A white precipitate of calcium oxalate, insoluble in ammonium oxalate and oxalic acid solution indicates the presence of oxalate ion.

$$CaCl_2 + Na_2C_2O_4 \longrightarrow \underset{\text{Calcium oxalate (White precipitate)}}{CaC_2O_4} + 2NaCl$$

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(b) $KMnO_4$ test

Filter the precipitate from test (a). Add dil. H_2SO_4 to it followed by dilute $KMnO_4$ solution and warm. Pink colour of $KMnO_4$ is discharged:

$$CaC_2O_4 + H_2SO_4 \longrightarrow \underset{\text{Calcium sulphate}}{CaSO_4} + \underset{\text{Oxalic acid}}{H_2C_2O_4}$$

$$2KMnO_4 + 3H_2SO_4 + 5H_2C_2O_4 \longrightarrow 2MnSO_4 + K_2SO_4 + 8H_2O + 10CO_2$$

Pass the gas evolved through lime water. A white precipitate is formed which dissolves on passing the gas for some more time.

7. Test of Borate ion ($BO_3^{3\ominus}$):

Various tests carried out for identification of borate radical are listed as follows.

1. **Solubility**

   a. The borates of alkali metals are soluble in water.

   b. The borates of other metals are sparingly soluble in water.

2. **Heating effects**

$$\underset{\text{orthoboric acid}}{B(OH)_3} \xrightarrow{100^\circ C} \underset{\text{metaboric acid}}{HBO_2} \xrightarrow{140^\circ C} \underset{\text{pyroboric acid}}{H_2B_4O_7} \xrightarrow{>500^\circ C} \underset{\text{anhydride of boric acid}}{B_2O_3}$$

Note: All kinds of borate solutions are alkaline in nature due to their hydrolysis.

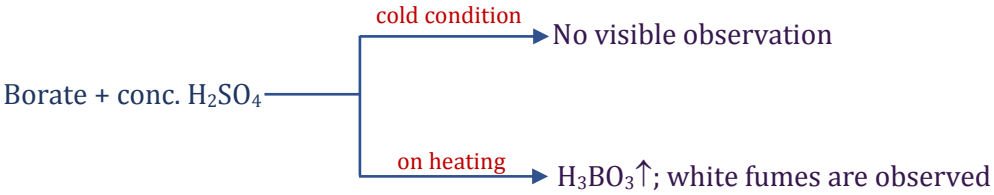
Orthoborate $\quad BO_3^{3\ominus} + 3H_2O \rightleftharpoons B(OH)_3 + 3OH^\ominus$

Pyroborate $\quad B_4O_7^{2\ominus} + 7H_2O \rightleftharpoons 4B(OH)_3 + 2OH^\ominus$

Metaborate $\quad BO_2^\ominus + 2H_2O \rightleftharpoons B(OH)_3 + OH^\ominus$

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3. **Test with conc. H_2SO_4 and conc. HCl:** In this test, when conc. H_2SO_4 is added to the solution of borate in cold conditions, no visible change is observed. On heating, white fumes of H_3BO_3 are observed



However, the addition of conc. HCl to borate ions in cold conditions leads to the formation of white crystals of H_3BO_3, but conc.HCl in hot condition cannot be used because it fumes a lot (even at room temperature).

$$\text{Borate + conc. HCl} \xrightarrow{\text{cold condition}} H_3BO_3\downarrow\text{; white crystals}$$

Reason: The $B(OH)_3$ produced forms hydrogen bonds with conc. H_2SO_4 and does not aggregate together to be precipitated out. In case of HCl, no hydrogen bonding takes place and $B(OH)_3$ formed aggregates to form white crystals

4. **Test with conc. H_2SO_4 and alcohol (green flame test):** When conc. H_2SO_4 is added to solution of borate ions in presence of methanol or ethanol, colourless fumes are evolved which burn with a green flame.

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Reaction involved :

$$B(OH)_3 + \left.\begin{matrix}3MeOH\\ or\\ 3EtOH\end{matrix}\right\} \xrightarrow{conc.H_2SO_4} \begin{matrix}B(OMe)_3\uparrow + 3H_2O\\ or\\ B(OEt_3)\uparrow + 3H_2O\end{matrix}$$

Note:

(i) Use of methyl alcohol is preferred due to high volatility of the product formed, i.e. $B(OMe)_3$.

(ii) Copper and barium salts also give similar green flame. To avoid their interference in the test, the following modification to the test is helpful to identify borate radical in presence of Cu and Ba salts. It is based upon the greater volatility of BF_3.

Thoroughly mix the borate with powdered CaF_2 and a little conc. H_2SO_4, and bring a little paste formed with the help of glass rod very close to the edge of the base of Bunsen flame without actually touching it. Volatile BF_3 is formed and produces green flame.

$$Na_2B_4O_7 + 6CaF_2 + 7\,H_2SO \longrightarrow 4BF_3\uparrow + 6CaSO_4 + Na_2SO_4 + 7H_2O$$

5. **Test with $AgNO_3$ solution:** White precipitate of $AgBO_2$ is formed from fairly concentrated borax solution, and on boiling, it turns brownish black due to the formation of Ag_2O.
Reactions involved:

$$B_4O_7^{2\ominus} + 4Ag^{\oplus} + H_2O \longrightarrow 4AgBO_2\downarrow + 2H^{\oplus}$$

$$2AgBO_2\downarrow + 3H_2O \xrightarrow{\Delta} Ag_2O\downarrow + 2H_3BO_3$$

6. **Test with $BaCl_2/CaCl_2/SrCl_2$ solution:** White precipitate of $M(BO_2)_2$ is obtained on addition of these salts to borate radical.

$$B_4O_7^{2\ominus} + 2M^{2\oplus} + H_2O \longrightarrow \underset{\text{white ppt.}}{2M(BO_2)_2\downarrow} + 2H^{\oplus}$$

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7. **Action of heat :**

$$\left.\begin{matrix}\text{Borax}\\ \text{Powder}\end{matrix}\right\} \xrightarrow[\text{the help of Pt wire loop}]{\text{heated with}} \text{Initially swells up and finally contracts into transparent glassy bead}$$

Reaction involved:

$$Na_2B_4O_7.10H_2O \xrightarrow{-10\,H_2O} \underbrace{2NaBO_2 + B_2O_3}_{\text{transparent glassy bead}}$$

The glass possesses the property of dissolving many oxides on heating, forming metaborates, which often have characteristic colours; and this is the basis of borax bead test for various metals.

8. **Bromothymol blue test:** Boric acid is a weak monobasic acid ($Ka = 5.8 \times 10^{-10}$) but on addition of certain organic polyhydroxy compounds, such as mannitol, glycerol, dextrose or invert sugar, it is transformed into a relatively strong acid, as shown in the following reactions.

$$H_3BO_3 + H_2O \rightleftharpoons [B(OH)_4]^{-\ominus} + H^{\oplus}$$



The formation of chelated complex in the second reaction helps shift the first reaction more towards right and boric acid becomes more strongly acidic.

Note: Ethylene glycol cannot give this test (reason is uncertain).

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Class - B:

Test for Sulphate and Phosphate

If no positive test is obtained in Steps-I and II, then tests for the presence of sulphate and phosphate ions are performed. These tests are summarised in Table 5.

Table 5 : Confirmatory tests for Sulphate and Phosphate

Ion	Confirmatory Test
Sulphate ($SO_4^{2\ominus}$)	(a) Take 1 mL water extract of the salt in water or sodium carbonate and after acidifying with dilute hydrochloric acid add $BaCl_2$ solution. White precipitate insoluble in conc. HCl or conc. HNO_3 is obtained. (b) Acidify the aqueous solution or sodium carbonate extract with acetic acid and add lead acetate solution. Appearance of white precipitate confirms the presence of $SO_4^{2\ominus}$ ion.
Phosphate ($PO_4^{3\ominus}$)	(a) Acidify sodium carbonate extract or the solution of the salt in water with conc. HNO_3 and add ammonium molybdate solution and heat to boiling. A canary yellow precipitate is formed.

1. Test of Sulphate ions [$SO_4^{2\ominus}$]

(a) Aqueous solution or sodium carbonate extract of the salt acidified with acetic acid on addition of barium chloride gives a white precipitate of barium sulphate insoluble in conc. HCl or conc. HNO_3.

$$Na_2SO_4 + BaCl_2 \longrightarrow \underset{\substack{\text{Barium sulphate}\\\text{(White precipitate)}}}{BaSO_4} + 2NaCl$$

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(b) Sulphate ions give white precipitate of lead sulphate when aqueous solution or sodium carbonate extract neutralised with acetic acid is treated with lead acetate solution.

$$Na_2SO_4 + (CH_3COO)_2Pb \longrightarrow \underset{\substack{\text{Lead sulphate}\\\text{(White precipitate)}}}{PbSO_4} + 2CH_3COONa$$

2. Test for Phosphate ion [$PO_4^{3\ominus}$]

(a) Add conc. HNO_3 and ammonium molybdate solution to the test solution containing phosphate ions and boil. A yellow colouration in solution or a canary yellow precipitate of ammonium-phosphomolybdate, $(NH_4)_3[P(Mo_3O_{10})_4]$ is formed. Each oxygen of phosphate has been replaced by Mo_3O_{10} group.

$$Na_2HPO_4 + 12(NH_4)_2MoO_4 + 23HNO_3 \longrightarrow \underset{\substack{\text{Canary yellow}\\\text{precipitate}}}{(NH_4)_3[P(Mo_3O_{10})_4]} + 2NaNO_3 + 21NH_4NO_3 + 12H_2O$$

Group Analysis:

(I) Analysis of Zero group cation ($NH_4^{\oplus}$ ion)

(a) Take 0.1 g of salt in a test tube and add 1-2 mL of NaOH solution to it and heat. If there is a smell of ammonia, this indicates the presence of ammonium ions. Bring a glass rod dipped in hydrochloric acid near the mouth of the test tube. White fumes are observed.

(b) Pass the gas through Nessler's reagent. Brown precipitate is obtained.

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Chemistry of Confirmatory Tests for $NH_4^{\oplus}$ ion

(a) Ammonia gas evolved by the action of sodium hydroxide on ammonium salts reacts with hydrochloric acid to give ammonium chloride, which is visible as dense white fume.

$(NH_4)_2SO_4 + 2NaOH \longrightarrow Na_2SO_4 + 2NH_3 + 2H_2O$

$NH_3 + HCl \longrightarrow NH_4Cl$

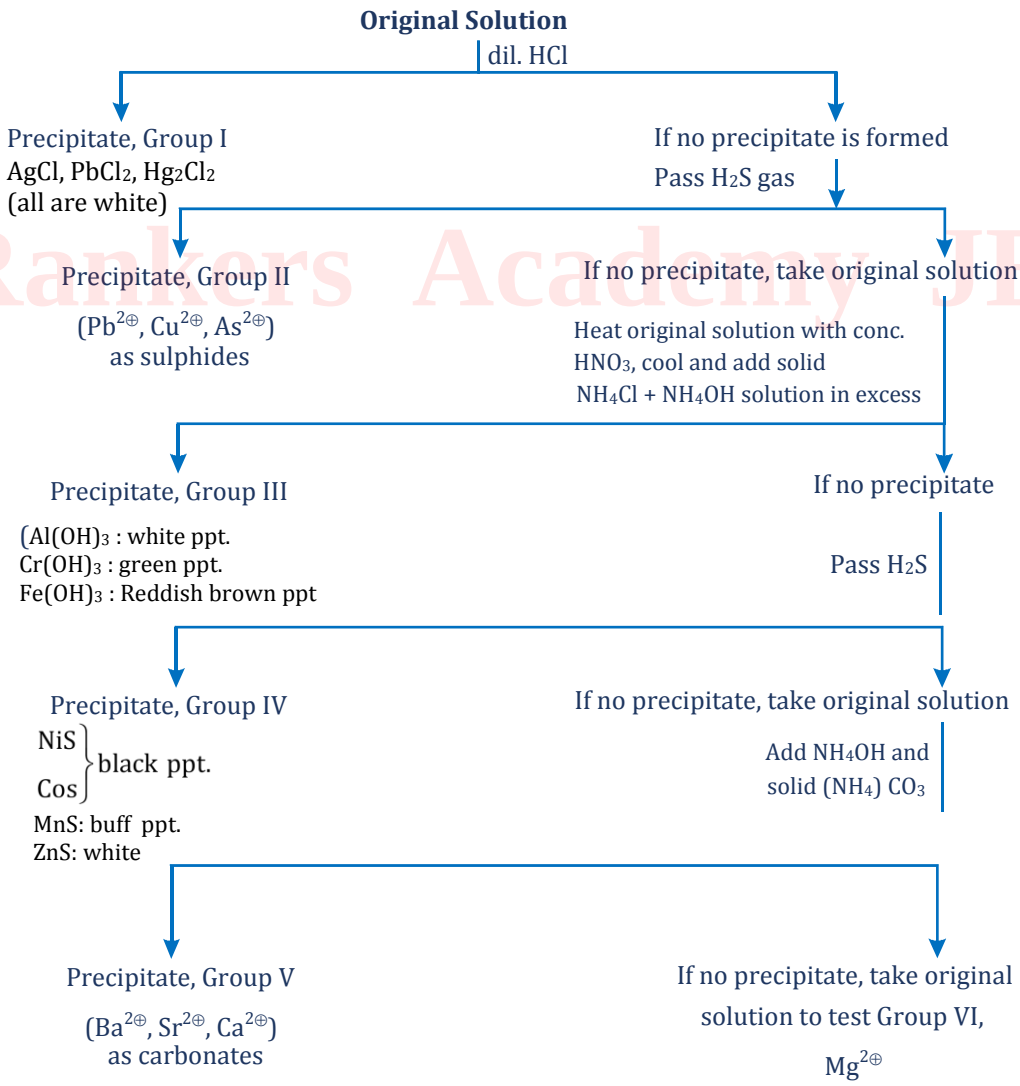
On passing the gas through Nessler's reagent, a brown colouration or a precipitate of basic mercury(II) amido-iodine is formed.

$$2K_2[HgI_4] + NH_3 + 3KOH \longrightarrow \underset{\text{Basic mercury (II) amido-iodine (Brown precipitate)}}{HgO.Hg(NH_2)I} + 7KI + 2H_2O$$

For the analysis of cations belonging to groups I-VI, the cations are precipitated from the original solution by using the group reagents (see Table) according to the scheme shown in the flow chart given below:

The separation of all the six groups is represented as below :

Flow Chart*



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Table 6 : Group reagents for precipitating ions

Group	Cations*	Group Reagent
Group zero	$NH_4^{\oplus}$	None
Group – I	$Ag^{\oplus}$, $Hg_2^{2\oplus}$, $Pb^{2\oplus}$	Dilute HCl
Group – II	(A) $Pb^{2\oplus}$, $Cu^{2\oplus}$, $Bi^{3\oplus}$, $Hg^{2\oplus}$, $Cd^{2\oplus}$,	H_2S gas in presence of dil. HCl
	(B) $As^{3\oplus/5\oplus}$, $Sb^{3\oplus/5\oplus}$, $Sn^{2\oplus}$, $Sn^{4\oplus}$	
Group – III	$Al^{3\oplus}$, $Fe^{3\oplus}$, $Cr^{3\oplus}$	NH_4OH in presence of NH_4Cl
Group – IV	$Co^{2\oplus}$, $Ni^{2\oplus}$, $Mn^{2\oplus}$, $Zn^{2\oplus}$	H_2S in presence of NH_4OH
Group – V	$Ba^{2\oplus}$, $Sr^{2\oplus}$, $Ca^{2\oplus}$	$(NH_4)_2CO_3$ in presence of NH_4OH
Group – VI	$Mg^{2\oplus}$	None

(II) Analysis of Group-I cations

Take a small amount of original solution (if prepared in hot conc. HCl) in a test tube and add cold water to it and cool the test tube under tap water. If a white precipitate appears, this indicates the presence of Pb^{2+} ions in group –I. On the other hand, if the original solution is prepared in water and on addition of dil. HCl, a white precipitate appears, this may also be Pb^{2+}. Confirmatory tests are described below in Table 7.

Table 7 : Confirmatory tests for Group-I cation ($Pb^{2\oplus}$)

Experiment	Observation
Dissolve the precipiate in hot water and divide the hot solution into three parts,	
1. Add potassium idoide solution to the first part.	A yellow precipiate is obtained.
2. To the second part add potassium chromate solution.	A yellow precipitate is obtained which is soluble, in NaOH and insoluble in ammonium acetate solution.
3. To the third part of the hot solution add few drops of alcohol and dilute sulphuric acid.	A white precipitate is obtained which is soluble in ammonium acetate solution.

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1. Tests of $Pb^{2\oplus}$ ions

Lead is precipitated as lead chloride in the first group. The precipitate is soluble in hot water.

- On adding potassium iodide (KI) solution, a yellow precipitate of lead iodide is obtained which confirms the presence of $Pb^{2\oplus}$ ions.

$$\underset{\text{(Hot solution)}}{PbCl_2} + 2KI \longrightarrow \underset{\text{Yellow precipitate}}{PbI_2} + 2KCl$$

This yellow precipitate (PbI_2) is soluble in boiling water and reappears on cooling as shining crystals.

- On addition of potassium chromate (K_2CrO_4) solution a yellow precipitate of lead chromate is obtained. This confirms the presence of $Pb^{2\oplus}$ ions.

$$\underset{\text{(Hot solution)}}{PbCl_2} + K_2CrO_4 \longrightarrow \underset{\text{Lead chromate (Yellow precipitate)}}{PbCrO_4} + 2KCl$$

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The yellow precipitate ($PbCrO_4$) is soluble in hot NaOH solution.

$$PbCrO_4 + 4NaOH \longrightarrow \underset{\text{Sodium tetra hydroxoplumbate (II)}}{Na_2[Pb(OH)_4]} + Na_2CrO_4$$

- A white precipitate of lead sulphate ($PbSO_4$) is formed on addition of alcohol followed by dil. H_2SO_4.

$$PbCl_2 + H_2SO_4 \longrightarrow \underset{\text{Lead sulphate (White precipitate)}}{PbSO_4} + 2HCl$$

Lead sulphate is soluble in ammonium acetate solution due to the formation of tetraacetatoplumbate(II) ions. This reaction may be promoted by addition of few drops of acetic acid.

$$PbSO_4 + 4CH_3COONH_4 \longrightarrow \underset{\text{Ammonium tetraacetatoplumbate(II)}}{(NH_4)_2[Pb(CH_3COO)_4]} + (NH_4)_2SO_4$$

2. Test of $Ag^{\oplus}$ ion:

- **Dilute hydrochloric acid/soluble chlorides :** White precipitate is formed.

$$Ag^{\oplus} + HCl \longrightarrow AgCl\downarrow + H^{\oplus}$$

The precipitate obtained after filtration is soluble in concentrated HCl.

$$AgCl\downarrow + Cl^{\ominus} \rightleftharpoons [AgCl_2]^{\ominus}$$

On dilution with water, the equilibrium shifts back to the left and the precipitate reappears. Dilute ammonia solution dissolves the precipitate forming a soluble complex.

$$AgCl\downarrow + 2NH_3 \rightleftharpoons [Ag(NH_3)_2]^{\oplus} + Cl^{\ominus}$$

Dilute nitric acid or hydrochloric acid neutralizes the excess ammonia and the precipitate reappears because the equilibrium is shifted backwards.

$$[Ag(NH_3)_2]Cl + 2HNO_3 \longrightarrow AgCl\downarrow (white) + 2NH_4NO_3.$$

- **Potassium iodide solution:** A bright yellow precipitate is formed which is insoluble in dilute ammonia but partially soluble in concentrated ammonia.

$$Ag^{\oplus} + I^{\ominus} \longrightarrow AgI\downarrow$$

The yellow precipitate is soluble in KCN and in $Na_2S_2O_3$.

$$AgI\downarrow + 2CN^{\ominus} \longrightarrow [Ag(CN)_2]^{\ominus} + I^{\ominus}\ ;\quad AgI\downarrow + 2S_2O_3^{2\ominus} \longrightarrow [Ag(S_2O_3)_2]^{3\ominus} + I^{\ominus}$$

- **Potassium chromate solution:** Red precipitate is formed which is soluble in dilute HNO_3 and in ammonia solution.

$$2Ag^{\oplus} + CrO_4^{2\ominus} \longrightarrow Ag_2CrO_4\downarrow$$

$$2Ag_2CrO_4\downarrow + 2H^{\oplus} \rightleftharpoons 4Ag^{\oplus} + Cr_2O_7^{2\ominus} + H_2O$$

$$2Ag_2CrO_4\downarrow + 4NH_3 \longrightarrow 2[Ag(NH_3)_2]^{\oplus} + CrO_4^{2\ominus}$$

- **Disodium hydrogen phosphate solution:** In neutral solution a yellow precipitate is formed with the reagent.

$$3Ag^{\oplus} + HPO_4^{2\ominus} \longrightarrow Ag_3PO_4\downarrow + H^{\oplus}$$

The yellow precipitate is soluble in nitric acid and ammonia solution.

- **Ammonia solution :** Brown precipitate is formed.

$$2Ag^{\oplus} + 2NH_3 + H_2O \longrightarrow Ag_2O\downarrow + 2NH_4^{\oplus}$$

Precipitate dissolves in ammonia.

$$Ag_2O\downarrow + 4NH_3 + H_2O \longrightarrow 2[Ag(NH_3)_2]^{\oplus} + 2OH^{\ominus}$$

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3. Test of $Hg_2^{2\oplus}$ ion:

- **Dilute HCl solution :** White precipitate is formed in cold solution.

$$Hg_2^{2\oplus} + 2HCl \longrightarrow Hg_2Cl_2\downarrow \text{(white)} + 2H^{\oplus}$$

- **Ammonia solution :** A mixture of mercury metal (black precipitate) and basic mercury (II) amido chloride (white precipitate) is formed.

$$2Hg_2Cl_2 + 4NH_4OH \longrightarrow \underbrace{HgO.Hg(NH_2)Cl\downarrow + Hg\downarrow}_{\text{black}} + 3NH_4Cl + 3H_2O$$

- **Dissolution of white precipitate (Hg_2Cl_2) in aquaregia :**

$$3Hg_2Cl_2 + 2HNO_3 + 6HCl \longrightarrow 6HgCl_2 + 2NO + 4H_2O$$

- **Potassium iodide solution :** A green precipitate is formed.

$$Hg_2^{2\oplus} + 2I^{\ominus} \longrightarrow Hg_2I_2\downarrow$$

Green precipitate in excess of reagent undergoes disproportionation reaction and a soluble $[HgI_4]^{2\ominus}$ ions and black mercury are formed.

$$Hg_2I_2\downarrow + 2I^{\ominus} \longrightarrow [HgI_4]^{2\ominus} + Hg\downarrow \text{(finely divided)}$$

Boiling the mercury (I) iodide precipitate with water, disproportionation takes place and a mixture of red mercury (II) iodide precipitate and black mercury is formed.

$$Hg_2I_2\downarrow \longrightarrow HgI_2\downarrow + Hg\downarrow$$

- **Potassium chromate solution :** A red crystalline precipitate is formed which turns black when solution of sodium hydroxide is added.

$$Hg_2^{2\oplus} + CrO_4^{2\ominus} \longrightarrow Hg_2CrO_4\downarrow ; Hg_2CrO_4\downarrow + 2OH^{\ominus} \longrightarrow Hg_2O\downarrow + CrO_4^{2\ominus} + H_2O$$

- **Potassium cyanide solution :** A black precipitate of mercury is obtained

$$Hg_2^{2\oplus} + 2CN^{\ominus} \longrightarrow Hg\downarrow + Hg(CN)_2 \text{ (soluble).}$$

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(III) Analysis of Group–II cations

If group-I is absent, add excess of water to the same test tube. Warm the solution and pass H_2S gas for 1-2 minutes (Fig. 1.6). Shake the test tube. If a precipitate appears, this indicates the presence of group-II cations. Pass more H_2S gas through the solution to ensure complete precipitation and separate the precipitate. If the colour of the precipitate is black, it indicates the presence of $Cu^{2\oplus}$ or $Pb^{2\oplus}$ ions.

If it is yellow in colour, then presence of $As^{3\oplus}$ ions is indicated.

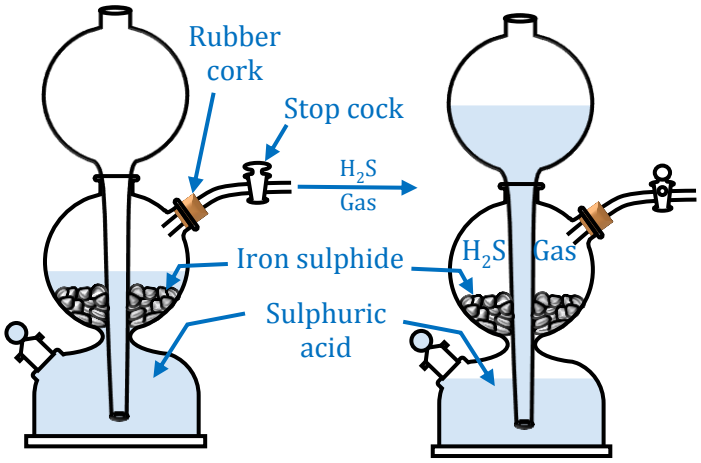


Fig. : Kipp's apparatus for preparation of H_2S gas

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Take the precipitate of group-II in a test tube and add excess of yellow ammonium sulphide solution to it. Shake the test tube. If the precipitate is insoluble, group II-A (copper group) is present. If the precipitate is soluble, this indicates the presence of group-II B (arsenic group). Confirmatory tests for the groups II A and II B are given in Table 8.

Table 8 : Confirmatory tests for the groups II A and II B cations

<table>
<tr><td colspan="2">Black precipitate of Group II A ions $Pb^{2\oplus}$, $Cu^{2\oplus}$ (insoluble in yellow ammonium sulphide is formed.</td><td>If a yellow precipitate soluble in yellow ammonium sulphide is formed then $As^{3\oplus}$ ion is present.</td></tr>
<tr><td colspan="2">Boil the precipitate of Group II A with dilute nitric acid and add a few drops of alcohol and dil. H_2SO_4.</td><td rowspan="2">Acidify this solution with dilute HCl. A yellow precipitate is formed. Heat the precipitate with concentrated nitric acid and add ammonium molybdate solution. A canary yellow precipitate is formed.</td></tr>
<tr><td>White precipitate confirms the presence of $Pb^{2\oplus}$ ions. Dissolve the precipitate in ammonium acetate solution. Acidify with acetic acid and divide the solution into two parts.
(i) To the first part add potassium chromate solution, a yellow precipitate is formed.
(ii) To the second part, add potassium iodide solution, a yellow precipitate is formed.</td><td>If no precipitate is formed, add excess of ammonium hydroxide solution. A blue solution is obtained, acidify it with acetic acid and add potassium ferrocyanide solution. A chocolate brown precipitate is formed.</td></tr>
</table>

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Group-II A (Copper Group)

Chemistry of confirmatory tests of Group-II A cations

1. Test for Lead ion ($Pb^{2\oplus}$) :

Lead sulphide precipitate dissolves in dilute HNO_3. On adding dil. H_2SO_4 and a few drops of alcohol to this solution a white precipitate of lead sulphate appears. This indicates the presence of lead ions.

$$3PbS + 8HNO_3 \longrightarrow 3Pb(NO_3)_2 + 2NO + 4H_2O + 3S$$

$$Pb(NO_3)_2 + H_2SO_4 \longrightarrow PbSO_4 + 2HNO_3$$

The white precipitate dissolves in ammonium acetate solution on boiling. When this solution is acidified with acetic acid and potassium chromate solution is added, a yellow precipitate of $PbCrO_4$ is formed. On adding potassium iodide solution, a yellow precipitate of lead iodide is formed.

$$PbSO_4 + 4CH_3COONH_4 \longrightarrow \underset{\text{Ammonium tetraacetatoplumbate(II)}}{(NH_4)_2[Pb(CH_3COO)_4]} + (NH_4)_2SO_4$$

$$Pb^{2\oplus} + CrO_4^{2\ominus} \longrightarrow \underset{\text{(Yellow precipitate)}}{PbCrO_4}\ ; \qquad Pb^{2\oplus} + 2I^{\ominus} \rightarrow \underset{\text{(Yellow precipitate)}}{PbI_2}$$

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2. Test for Copper ion ($Cu^{2\oplus}$) :

(a) Copper sulphide dissolves in nitric acid due to the formation of copper nitrate.

$$3CuS + 8HNO_3 \longrightarrow 3Cu(NO_3)_2 + 2NO + 3S + 4H_2O$$

On heating the reaction mixture for long time, sulphur is oxidised to sulphate and copper sulphate is formed and the solution turns blue. A small amount of NH_4OH precipitates basic copper sulphate which is soluble in excess of ammonium hydroxide due to the formation of tetraamminecopper (II) complex.

$$S + 2HNO_3 \longrightarrow H_2SO_4 + 2NO$$

$$2Cu^{2\oplus} + SO_4^{2\ominus} + 2NH_3 + 2H_2O \longrightarrow Cu(OH)_2.CuSO_4 + 2NH_4^{\oplus}$$

$$Cu(OH)_2.CuSO_4 + 8NH_3 \longrightarrow \underset{\substack{\text{Tetraamminecopper (II)}\\ \text{sulphate (Deep blue)}}}{2[Cu(NH_3)_4]SO_4} + 2OH^{\ominus} + SO_4^{2\ominus}$$

(b) The blue solution on acidification with acetic acid and then adding potassium ferrocyanide $K_4[Fe(CN)_6]$ solution gives a chocolate colouration due to the formation of copper ferrocyanide i.e. $Cu_2[Fe(CN)_6]$.

$$[Cu(NH_3)_4]SO_4 + 4CH_3COOH \longrightarrow CuSO_4 + 4CH_3COONH_4$$

$$2CuSO_4 + \underset{\substack{\text{Potassium}\\ \text{hexacyanoferrate (II)}}}{K_4[Fe(CN)_6]} \longrightarrow \underset{\substack{\text{Copper}\\ \text{hexacyanoferrate (II)}\\ \text{(Chocolate brown precipitate)}}}{Cu_2[Fe(CN)_6]} + 2K_2SO_4$$

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3. Test of Mercury (II) ion ($Hg^{2\oplus}$) :

- **Precipitation with H_2S in acidic medium :** Black precipitate is formed. Precipitate insoluble in water, hot dilute HNO_3 ,alkali hydroxides, or colourless ammonium sulphide.

$$Hg^{2\oplus} + H_2S \xrightarrow{H^{\oplus}} HgS\downarrow + 2H^{\oplus}$$

☞ Na_2S (2M) dissolves the precipitate forming soluble complex.

$$HgS + S^{2\ominus} \longrightarrow [HgS_2]^{2\ominus}$$

☞ Aqua regia dissolves the precipitate.

$$3\,HgS + 6\,HCl + 2HNO_3 \longrightarrow 3\,HgCl_2 + 3S\downarrow + 2NO\uparrow + 4\,H_2O$$

Under these circumstance $HgCl_2$ is undissociated. When solution is heated white precipitate of sulphur dissolves forming H_2SO_4.

$$2\,HNO_3 + S\downarrow \longrightarrow SO_4^{2\ominus} + 2\,H^{\oplus} + 2NO\uparrow$$

- **Stannous chloride solution :** When added in moderate amounts silky white precipitate is formed.

$$2HgCl_2 + SnCl_2 \longrightarrow SnCl_4 + Hg_2Cl_2\downarrow$$

If more reagent is added, Hg (I) chloride is reduced to black precipitate of mercury.

$$Hg_2Cl_2 + SnCl_2 \longrightarrow SnCl_4 + 2Hg\downarrow$$

- **Potassium iodide solution :** On slow addition red precipitate is formed.

$$Hg^{2\oplus} + 2I^{\ominus} \longrightarrow HgI_2\downarrow$$

Precipitate dissolves in excess of KI forming colourless soluble complex.

$$HgI_2 + 2I^{\ominus} \longrightarrow [HgI_4]^{2\ominus}$$

☞ KCN does not have any effect.

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- **Copper chips , sheet or coin :** A black precipitate of mercury is formed.

$$Hg^{2\oplus} + Cu \longrightarrow Hg\downarrow + Cu^{2\oplus}$$

- **Sodium hydroxide solution :** When added in small quantity brownish–red precipitate of varying composition is formed and in stoichiometric amounts precipitate turns to yellow when Hg (II) oxide is formed.

$$Hg^{2\oplus} + 2OH^{\ominus} \longrightarrow HgO\downarrow + H_2O$$

Precipitate is insoluble in excess reagent but dissolves readily in acids and this can be used to differentiate Hg (I) from Hg (II).

- **Ammonia solution :** White precipitate of mixed composition (Mercury (II) oxide + Mercury (II) amido nitrate) is formed with metal nitrate.

$$2\,Hg^{2\oplus} + NO_3^{\ominus} + 4NH_3 + H_2O \longrightarrow HgO.Hg(NH_2)NO_3\downarrow + 3\,NH_4^{\oplus}$$

- **Cobalt (II) thiocyanate test :** When reagent is added to an aqueous solution of $Hg^{2\oplus}$ ions and the walls of the test tube is stirred with a glass rod, deep–blue crystalline precipitate is formed.

$$Hg^{2\oplus} + Co^{2\oplus} + 4\,SCN^{\ominus} \longrightarrow Co^{2\oplus}[Hg(SCN)_4]^{2\ominus}\downarrow \text{ or } Hg[Co(NCS)_4]\downarrow$$

☞ In place of Cobalt (II) thiocyanate, $Co(CH_3COO)_2$ and NH_4SCN can be added to the aqueous solution of $Hg^{2\oplus}$ ions.

4. Test of Cadmium ion ($Cd^{2\oplus}$)

- **Precipitation with H_2S in acidic medium :** Yellow precipitate is formed which dissolves in hot dilute HNO_3.

$$Cd^{2\oplus} + H_2S \xrightarrow{H^{\oplus}} CdS\downarrow + 2H^{\oplus}$$

$$CdS + 8HNO_3 \longrightarrow 3Cd(NO_3)_2 + 4H_2O + 2NO + 3S$$

☞ Precipitate does not dissolve in KCN.

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- **Ammonia solution (Dropwise addition) :** Ammonium hydroxide first gives white precipitate of $Cd(OH)_2$ which gets dissolve in excess of reagent forming a soluble complex.

$$Cd^{2\oplus} + 2\,NH_3 + 2\,H_2O \rightleftharpoons Cd(OH)_2\downarrow + 2\,NH_4^{\oplus}$$

$$Cd(OH)_2\downarrow + 4\,NH_3 \longrightarrow [Cd(NH_3)_4]^{2\oplus} + 2\,OH^{\ominus}$$

- **Potassium cyanide :** Initially a white precipitate of $Cd(CN)_2$ is formed which in excess of reagent dissolves forming a soluble complex.

$$Cd^{2\oplus} + 2\,CN^{\ominus} \longrightarrow Cd(CN)_2\downarrow$$

$$Cd(CN)_2\downarrow + 2\,CN^{\ominus} \longrightarrow [Cd(CN)_4]^{2\ominus}$$

The colourless soluble complex is not too stable, therefore, reacts with H_2S gas forming a yellow precipitate of CdS.

$$[Cd(CN)_4]^{2\ominus} + H_2S \longrightarrow CdS\downarrow + 2\,H^{\oplus} + 4\,CN^{\ominus}$$

☞ KI forms no precipitate (distinction from Copper)

5. Test of Bismuth ion ($Bi^{3\oplus}$) :

- **Precipitation with H_2S in acidic medium :** Black precipitate is formed which is insoluble in cold dilute HNO_3 and yellow ammonium sulphide.

$$2Bi^{3\oplus} + 3H_2S \xrightarrow{H^{\oplus}} Bi_2S_3\downarrow \text{(black)} + 6H^{\oplus}$$

$$Bi_2S_3 + 8HNO_3 \longrightarrow 2Bi(NO_3)_3 + 2NO + 3S + 4H_2O$$

☞ $Bi_2S_3\downarrow + 6\,HCl$ (boiling concentrated) $\longrightarrow 2\,Bi^{3\oplus} + 6\,Cl^{\ominus} + 3\,H_2S$

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- **Sodium hydroxide :** White precipitate is formed with the reagent.

  $Bi^{3\oplus} + 3\ OH^{\ominus} \longrightarrow Bi(OH)_3\downarrow$

  Very slightly soluble in excess reagent in cold solution but soluble in acids.

  $Bi(OH)_3\downarrow + 3H^{\oplus} \longrightarrow Bi^{3\oplus} + H_2O$

  ☞ Precipitate on boiling loses water and turns yellowish white which is oxidised to $BiO_3^{\ominus}$ by H_2O_2.

  $Bi(OH)_3 \longrightarrow BiO.OH\downarrow + H_2O$

  $BiO.OH + H_2O_2 \longrightarrow BiO_3^{\ominus} + H^{\oplus} + H_2O$

- **Dilution with water :** Solution of bismuth salts gives white precipitate when water is added in larger quantity.

  $Bi^{3\oplus} + NO_3^{\ominus} + H_2O \longrightarrow BiO(NO_3)\downarrow + 2H^{\oplus}$

  $Bi^{3\oplus} + Cl^{\ominus} + H_2O \longrightarrow BiOCl\downarrow$ (bismuth oxychloride or bismuthyl chloride) $+ 2H^{\oplus}$

  Soluble in mineral acids (dilute) but insoluble in tartaric acid (distinction from antimony) and in alkali hydroxide (distinction from tin).

- **Potassium iodide :** When the reagent is added dropwise to a solution containing $Bi^{3\oplus}$ ions, a black precipitate is formed.

  $Bi^{3\oplus} + 3I^{\ominus} \longrightarrow BiI_3\downarrow$

  The precipitate dissolves in excess KI forming orange coloured soluble complex.

  $BiI_3 + I^{\ominus} \rightleftharpoons [BiI_4]^{\ominus}$

  On dilution the reaction is reversed and black BiI_3 is reprecipitated.

  ☞ On heating with water black precipitate of BiI_3 turns orange.

  $BiI_3\downarrow + H_2O \longrightarrow BiOI\downarrow + 2H^{\oplus} + 2I^{\ominus}$

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Group-II B (Arsenic Group)

If group- II precipitate dissolves in yellow ammonium sulphide and the colour of the solution is yellow, this indicates the presence of $As^{3\oplus}$ ions. Ammonium thioarsenide formed on dissolution of As_2S_3, decomposes with dil. HCl, and a yellow precipitate of arsenic (V) sulphide is formed which dissolves in concentrated nitric acid on heating due to the formation of arsenic acid. On adding ammonium molybdate solution to the reaction mixture and heating, a canary yellow precipitate is formed. This confirms the presence of $As^{3\oplus}$ ions.

$$\underset{}{As_2S_3} + \underset{\text{Yellow ammonium sulphide}}{3(NH_4)_2S} \longrightarrow 2(NH_4)_3AsS_4 + S$$

$$2(NH_4)_3AsS_4 + 6HCl \rightarrow As_2S_5 + 3H_2S + 6NH_4Cl$$

$$3As_2S_5 + 10HNO_3 + 4H_2O \longrightarrow \underset{\text{Arsenic acid}}{6H_3AsO_4} + 10NO + 15S$$

$$\underset{\text{Arsenic acid}}{H_3AsO_4} + \underset{\text{Ammonium molybdate}}{12(NH_4)_2MoO_4} + 21HNO_3 \rightarrow \underset{\substack{\text{Ammonium arsenomolybdate}\\ \text{(yellow precipitate)}}}{(NH_4)_3[As(Mo_3O_{10})_4]} + 21NH_4NO_3 + 12H_2O$$

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Test of $Sn^{2\oplus}$:

(a) Test by $H^{\oplus}/H_2S$:

$$Sn^{2\oplus} \xrightarrow{H^{\oplus}/H_2S} \underset{\text{(Brown)}}{SnS\downarrow}$$

(b) Test by NaOH :

$$Sn^{2\oplus} \xrightarrow{NaOH} \underset{\text{(White ppt)}}{Sn(OH)_2\downarrow} \xrightarrow{\text{excess}} Na_2[Sn(OH)_4]$$

(c) Test by NH_4OH :

$$Sn^{2\oplus} \xrightarrow{NH_4OH} \underset{\text{(White ppt)}}{Sn(OH)_2\downarrow} \xrightarrow{\text{excess}} \text{(Insoluble)}$$

(d) Test by KI (aq.) :

$$Sn^{2\oplus} \xrightarrow{KI\ (aq.)} \text{No ppt}$$

(e) Test by $HgCl_2$:

$$Sn^{2\oplus} \xrightarrow{HgCl_2} \underset{\text{(White)}}{Sn^{4\oplus} + Hg_2Cl_2\downarrow} \xrightarrow{\text{excess } Sn^{2\oplus}} \underset{\text{(Black)}}{2Hg\downarrow + SnCl_4}$$

(f) Test by Hg_2Cl_2 :

$$Sn^{2\oplus} \xrightarrow{Hg_2Cl_2} \underset{\text{(Black)}}{2Hg\downarrow + Sn^{4\oplus}}$$

(g) Test by $Fe^{3\oplus}/H^{\oplus}$:

$$Sn^{2\oplus} \xrightarrow[\text{(Yellow ppt)}]{Fe^{3\oplus}/H^{\oplus}} \underset{\text{(green)}}{Fe^{2\oplus} + Sn^{4\oplus}}$$

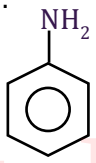
(h) Test by $K_2Cr_2O_7/H^{+}$:

$$Sn^{2\oplus} \xrightarrow{K_2Cr_2O_7/H^{\oplus}} Cr^{3\oplus} + Sn^{4\oplus}$$

(i) Test by $MnO_4^{\ominus}/H^{\oplus}$

$$Sn^{2\oplus} \xrightarrow{MnO_4^{\ominus}/H^{\oplus}} Mn^{2\oplus} + Sn^{4\oplus}$$

(j) Test by $C_6H_5NO_2/HCl$:

$$Sn^{2\oplus} \xrightarrow{C_6H_5NO_2/HCl} \text{  }$$

(k) Test by Excess KOH + $Bi(NO_3)_3$

$$Sn^{2\oplus} \xrightarrow[KOH + Bi(NO_3)_3]{\text{Excess}} Sn^{4\oplus} + \underset{\text{(Black)}}{Bi\downarrow}$$

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(IV) Analysis of Group–III cations

If group-II is absent, take original solution and add 2-3 drops of conc. HNO_3 to oxidise $Fe^{2\oplus}$ ions to $Fe^{3\oplus}$ ions. Heat the solution for a few minutes. After cooling add a small amount of solid ammonium chloride (NH_4Cl) and an excess of ammonium hydroxide (NH_4OH) solution till it smells of ammonia. Shake the test tube. If a brown or white precipitate is formed, this indicates the presence of group-III cations. Confirmatory tests of group-III cations are summarised in Table 9.

Observe the colour and the nature of the precipitate. A gelatinous white precipitate indicates the presence of aluminium ion ($Al^{3\oplus}$). If the precipitate is brown in colour, this indicates the presence of ferric ions ($Fe^{3\oplus}$).

Table 9 : Confirmatory test for Group-III cations

Brown precipitate $Fe^{3\oplus}$		**White precipitate $Al^{3\oplus}$**	
	Dissolve the precipitate in dilute HCl and divide the solution into two parts.		Dissolve the white precipitate in dilute HCl and divide into two parts.
(a)	To the first part add potassium ferrocyanide solution [Potassium hexacyanoferrate (II)]. A blue precipitate/colouration appears.	(a)	To the first part add sodium hydroxide solution and warm. A white gelatinous precipitate soluble in excess of sodium hydroxide solution is obtained
(b)	To the second part add potassium thiocyanate solution. A blood red colouration appears.	(b)	To the second part first add blue litmus solution and then ammonium hydroxide solution drop by drop along the sides of the test tube. A blue floating mass in the colourless solution is obtained.

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Chemistry of confirmatory tests of Group - III cations

When original solution is heated with concentrated nitric acid, ferrous ion are oxidised to ferric ions.

$$2FeCl_2 + 2HCl + [O] \longrightarrow 2FeCl_3 + H_2O$$

Their group cations are precipitated as their hydroxides, which dissolve in dilute hydrochloric acid due to the formation of corresponding chlorides.

1. Test for Aluminium ions ($Al^{3\oplus}$)

(a) When the solution containing aluminium chloride is treated with sodium hydroxide a white gelationus precipitate of aluminium hydroxide is formed which is soluble in excess of sodium hydroxide solution due to the formation of sodium meta aluminate.

$$AlCl_3 + 3NaOH \longrightarrow Al(OH)_3 + 3NaCl$$

$$\underset{\text{White gelatinous precipitate}}{Al(OH)_3} + NaOH \longrightarrow \underset{\text{Sodium meta aluminate}}{NaAlO_2} + 2H_2O$$

(b) In the second test when blue litmus is added to the solution, a red colouration is obtained due to the acidic nature of the solution. On addition of NH_4OH solution drop by drop, the solution becomes alkaline and aluminium hydroxide is precipitated. Aluminium hydroxide adsorbs blue colour from the solution and forms insoluble adsorption complex named 'lake'. Thus, a blue mass floating in the colourless solution is obtained. The test is therefore called **lake test.**

2. Test for ferric ions ($Fe^{3\oplus}$)

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Reddish brown precipitate of ferric hydroxide dissolves in hydrochloric acid and ferric chloride is formed.

$$Fe(OH)_3 + 3HCl \longrightarrow FeCl_3 + 3H_2O$$

(a) When the solution containing ferric chloride is treated with potassium ferrocyanide solution a blue precipitate/colouration is obtained. The colour of the precipitate is Prussian blue. It is ferric ferro-cyanide. The reaction takes place as follows:

$$4FeCl_3 + \underset{\text{Potassium ferrocyanide}}{3K_4[Fe(CN)_6]} \longrightarrow \underset{\text{Prussian blue precipitate}}{Fe_4[Fe(CN)_6]_3} + 12KCl$$

If potassium hexacyanoferrate (II) (i.e. potassium ferrocyanide) is added in excess then a product of composition $KFe[Fe(CN)_6]$ is formed. This tends to form a colloidal solution ('soluble Prussian blue') and cannot be filtered.

$$FeCl_3 + K_4[Fe(CN)_6] \longrightarrow \underset{\text{(Soluble prussian blue)}}{KFe[Fe(CN)_6]} + 3KCl$$

(b) To the second part of the solution, add potassium thiocyanate (potassium sulphocyanide) solution. The appearance of a blood red colouration confirms the presence of $Fe^{3\oplus}$ ions.

$$Fe^{3\oplus} + SCN^{\ominus} \longrightarrow \underset{\text{Blood red colour}}{[Fe(SCN)]^{3\oplus}}$$

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(V) Analysis of group-IV cations

If group-III is absent, pass H_2S gas in the solution of group-III for a few minutes. If a precipitate appears (white, black or flesh coloured), this indicates the presence of group-IV cations. Table 10 gives a summary of confirmatory tests of group-IV cations.

Table 10 : Confirmatory test for Group - IV cations

White precipitate $(Zn)^{2\oplus}$	Flesh coloured precipitate $(Mn)^{2\oplus}$	Black precipitate $(Ni^{2\oplus}, Co^{2\oplus})$
Dissolve the precipitate in dilute HCl by boiling. Divide the solution into two parts. (a) To the first part add sodium hydroxide solution. A white precipitate soluble in excess of sodium hydroxide solution confirms the presence of $Zn^{2\oplus}$ ions. (b) Neutralise the second part with a mmonium hydroxide solution and add potassium ferrocyanide solution. A bluish white precipitate appears	Dissolve the precipitate in dilute HCl by boiling, then add sodium hydroxide solution in excess. A white precipitate is formed which turns brown on keeping.	Dissolve the precipitate in aqua regia. Heat the solution to dryness and cool. Dissolve the residue in water and divide the solution into two parts. (a) To the first part of the solution add ammonium hydroxide solution till it becomes alkaline. Add a few drops of dimethyl glyoxime and shake the test tube. Formation of a bright red precipitate confirms the presence of Ni^{2+} ions. (b) Neutralise the second part with ammonium hydroxide solution. Acidify it with dilute acetic acid and add solid potassium nitrite. A yellow precipitate confirms the presence of Co^{2+} ions.

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Chemistry of confirmatory tests of Group–IV cations

Fourth group cations are precipitated as their sulphides. Observe the colour of the precipitate. A white colour of the precipitate indicates the presence of zinc ions, a flesh colour indicates the presence of manganese ions and a black colour indicates the presence of $Ni^{2\oplus}$ or $Co^{2\oplus}$ ions.

1. Test for Zinc ion $(Zn^{2\oplus})$

Zinc sulphide dissolves in hydrochloric acid to form zinc chloride.

$$ZnS + 2HCl \longrightarrow ZnCl_2 + H_2S$$

(a) On addition of sodium hydroxide solution, it gives a white precipitate of zinc hydroxide, which is soluble in excess of NaOH solution on heating. This confirms the presence of $Zn^{2\oplus}$ ions.

$$ZnCl_2 + 2NaOH \longrightarrow Zn(OH)_2 + 2NaCl$$

$$Zn(OH)_2 + 2NaOH \longrightarrow \underset{\text{Sodium zincate}}{Na_2ZnO_2} + 2H_2O$$

(b) When potassium ferrocyanide $K_4[Fe(CN)_6]$ solution is added to the solution after neutralisation by NH_4OH solution, a white or a bluish white precipitate of zinc ferrocyanide appears.

$$2ZnCl_2 + K_4[Fe(CN)_6] \longrightarrow \underset{\text{Zinc Ferrocyanide}}{Zn_2[Fe(CN)_6]} + 4KCl$$

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2. Test for Manganese ion ($Mn^{2\oplus}$)

Manganese sulphide precipitate dissolves in dil. HCl on boiling. On addition of NaOH solution in excess, a white precipitate of manganese hydroxide is formed which turns brown due to atmospheric oxidation into hydrated manganese dioxide.

$$MnS + 2HCl \longrightarrow MnCl_2 + H_2S$$

$$MnCl_2 + 2NaOH \longrightarrow \underset{\text{(White precipitate)}}{Mn(OH)_2} + 2NaCl$$

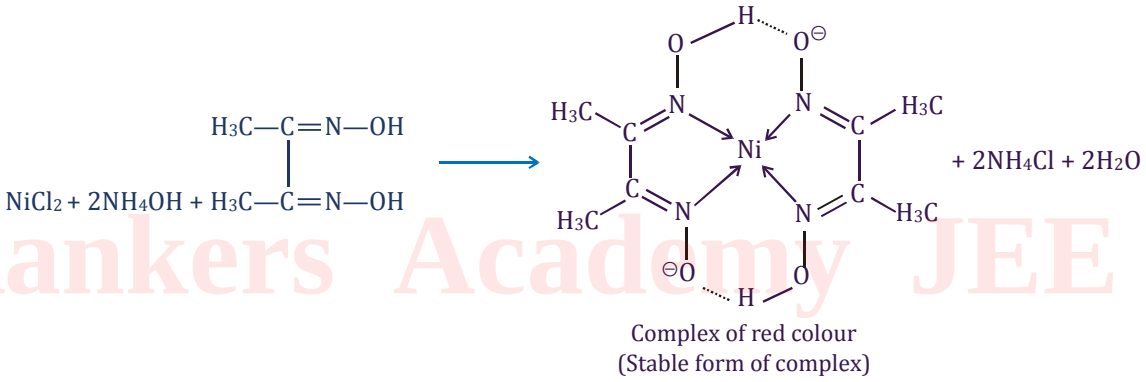
$$Mn(OH)_2 + [O] \longrightarrow \underset{\substack{\text{Hydrated manganese dioxide}\\ \text{(Brown colour)}}}{MnO(OH)_2}$$

3. Test for Nickel ion ($Ni^{2\oplus}$)

The black precipitate of nickel sulphide dissolves in aqua regia and the reaction takes place as follows:

$$3NiS + 2HNO_3 + 6HCl \longrightarrow 3NiCl_2 + 2NO + 3S + 4H_2O$$

After treatment with aqua regia nickel-chloride is obtained which is soluble in water. When dimethyl glyoxime is added to the aqueous solution of nickel chloride, made alkaline, by adding NH_4OH solution, a brilliant red precipitate is obtained.



Complex of red colour
(Stable form of complex)

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4. Test for Cobalt ion ($Co^{2\oplus}$)

Cobalt sulphide dissolves in aqua regia in the same manner as nickel sulphide. When the aqueous solution of the residue obtained after treatment with aqua regia is treated with a strong solution of potassium nitrite after neutralisation with ammonium hydroxide and the solution is acidified with dil. acetic acid, a yellow precipitate of the complex of cobalt named potassium hexanitritocobaltate (III) is formed.

$$CoS + HNO_3 + 3HCl \longrightarrow CoCl_2 + NOCl + S + 2H_2O$$

$$CoCl_2 + 7KNO_2 + 2CH_3COOH \longrightarrow \underset{\substack{\text{Potassium}\\ \text{hexanitritocobaltate(III)}\\ \text{(Yellow precipitate)}}}{K_3[Co(NO_2)_6]} + 2KCl + 2CH_3COOK + NO + H_2O$$

(VI) Analysis of Group–V cations

If group-IV is absent then take original solution and add a small amount of solid NH_4Cl and an excess of NH_4OH solution followed by solid ammonium carbonate $(NH_4)_2CO_3$. If a white precipitate appears, this indicates the presence of group–V cations.

Dissolve the white precipitate by boiling with dilute acetic acid and divide the solution into three parts one each for $Ba^{2\oplus}$, $Sr^{2\oplus}$ and $Ca^{2\oplus}$ ions. Preserve a small amount of the precipitate for flame test. Summary of confirmatory tests is given in Table 11.

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Table 11 : Confirmatory test for Group - V cations

Dissolve the precipitate by boiling with dilute acetic acid and divide the solution into three parts one each for $Ba^{2\oplus}$, $Sr^{2\oplus}$ and $Ca^{2\oplus}$ ions		
$Ba^{2\oplus}$ ions	**$Sr^{2\oplus}$ ions**	**$Ca^{2\oplus}$ ions**
(a) To the first part add potassium chromate solution. A yellow precipitate appears.	(a) If barium is absent, take second part of the solution and add ammonium sulphate solution. Heat and scratch the sides of the test tub e with a glass rod and cool. A white precipitate is formed.	(a) If both barium and strontium are absent, take the third part of the solution. Add ammonium oxalate solution and shake well. A white precipitate of calcium oxalate is obtained.
(b) Perform the flame test with the preserved precipitate. A grassy green flame is obtained.	(b) Perform the flame test with the preserved precipitate. A crimson-red flame confirms the presence of $Sr^{2\oplus}$ ions.	(b) Perform the flame test with the preserved precipitate. A brick red flame, which looks greenish- yellow through blue glass, confirms the presence of $Ca^{2\oplus}$ ions.

Chemistry of Confirmatory Tests of Group–V cations

The Group–V cations are precipitated as their carbonates which dissolve in acetic acid due to the formation of corresponding acetates.

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1. **Test for Barium ion ($Ba^{2\oplus}$)**

(a) Potassium chromate (K_2CrO_4) solution gives a yellow precipitate of barium chromate when the solution of fifth group precipitate in acetic acid is treated with it.

$$BaCO_3 + 2CH_3COOH \longrightarrow (CH_3COO)_2\,Ba + H_2O + CO_2$$

$$(CH_3COO)_2Ba + K_2CrO_4 \longrightarrow \underset{\text{Barium chromate (yellow precipitate)}}{BaCrO_4} + 2CH_3COOK$$

(b) **Flame test :** Take a platinum wire and dip it in conc. HCl. Heat it strongly until the wire does not impart any colour to the non-luminous flame. Now dip the wire in the paste of the (Group–V) precipitate in conc. HCl. Heat it in the flame. A grassy green colour of the flame confirms the presence of $Ba^{2\oplus}$ ions.

2. **Test for Strontium ion ($Sr^{2\oplus}$)**

(a) Solution of V group precipitate in acetic acid gives a white precipitate of strontium sulphate with ammonium sulphate solution on heating and scratching the sides of the test tube with a glass rod.

$$SrCO_3 + 2CH_3COOH \longrightarrow (CH_3COO)_2Sr + H_2O + CO_2$$

$$(CH_3COO)_2\,Sr + (NH_4)_2SO_4 \longrightarrow \underset{\text{Strontium sulphate (White precipitate)}}{SrSO_4} + 2CH_3COONH_4$$

(b) **Flame test :** Perform the flame test as given in the case of $Ba^{2\oplus}$. A crimson red flame confirms the presence of $Sr^{2\oplus}$ ions.

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3. Test for Calcium ion ($Ca^{2\oplus}$)

(a) Solution of the fifth group precipitate in acetic acid gives a white precipitate with ammonium oxalate solution.

$$CaCO_3 + 2CH_3COOH \longrightarrow (CH_3COO)_2Ca + H_2O + CO_2$$

$$(CH_3COO)_2Ca + \underset{\text{Ammonium oxalate}}{(NH_4)_2C_2O_4} \longrightarrow \underset{\text{Calcium oxalate (White precipitate)}}{(COO)_2Ca} + 2CH_3COONH_4$$

(b) **Flame test :** Perform the flame test as mentioned above. Calcium imparts brick red colour to the flame which looks greenish-yellow through blue glass.

(VII) Analysis of Group–VI cations

If group–V is absent then perform the test for $Mg^{2\oplus}$ ions as given below.

Chemistry of Confirmatory Tests of Group–VI cations

Test for Magnesium ion ($Mg^{2\oplus}$)

(a) If group–V is absent then the solution may contain magnesium carbonate, which is soluble in water in the presence of ammonium salts because the equilibrium is shifted towards the right hand side.

$$NH_4^{\oplus} + CO_3^{2\ominus} \longrightarrow NH_3 + HCO_3^{\ominus}$$

The concentration of carbonate ions required to produce a precipitate is not attained. When disodium hydrogen phosphate solution is added and the inner walls of the test tube are scratched with a glass rod, a white crystalline precipitate of magnesium ammonium phosphate is formed which indicates the presence of $Mg^{2\oplus}$ ions.

$$Mg^{2\oplus} + Na_2HPO_4 \longrightarrow \underset{\text{Magnesium ammonium phosphate (White recipitate)}}{Mg(NH_4)PO_4} + NH_4OH + 2Na^{\oplus} + H_2O$$

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Note down the observations and the inferences of the qualitative analysis in tabular form as given in the specimen record.

Heating Devices:

Heating during the laboratory work can be done with the help of a gas burner, spirit lamp or a kerosene lamp. The gas burner used in the laboratory is usually Bunsen burner. Various parts of Bunsen burner are shown in Fig. The description of these parts is as follows :

(A) Parts of Bunsen Burner

1. The Base

Heavy metallic base is connected to a side tube called gas tube. Gas from the source enters the burner through the gas tube and passes through a small hole called Nipple or Nozzle and enters into the burner tube under increased pressure and can be burnt at the upper end of the burner tube.

2. The Burner Tube

It is a long metallic tube having two holes diametrically opposite to each other near the lower end which form the air vent. The tube can be screwed at the base. The gas coming from the nozzle mixes with the air coming through the air vent and burns at its upper end.

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3. The Air Regulator

It is a short metallic cylindrical sleeve with two holes diametrically opposite to each other. When it is fitted to the burner tube, it surrounds the air vent of the burner tube. To control the flow of air through the air vent, size of its hole is adjusted by rotating the sleeve.

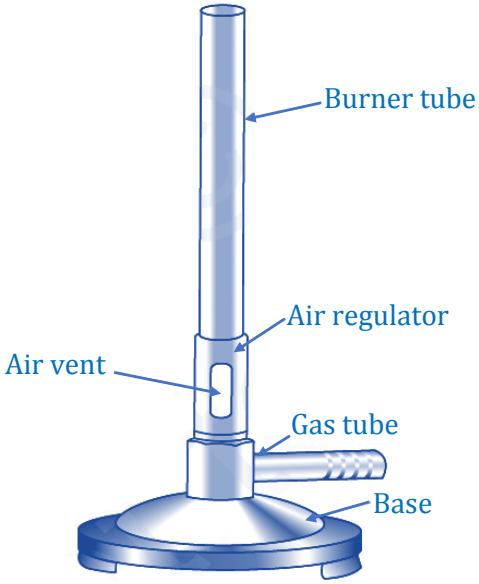


Fig. : Bunsen burner

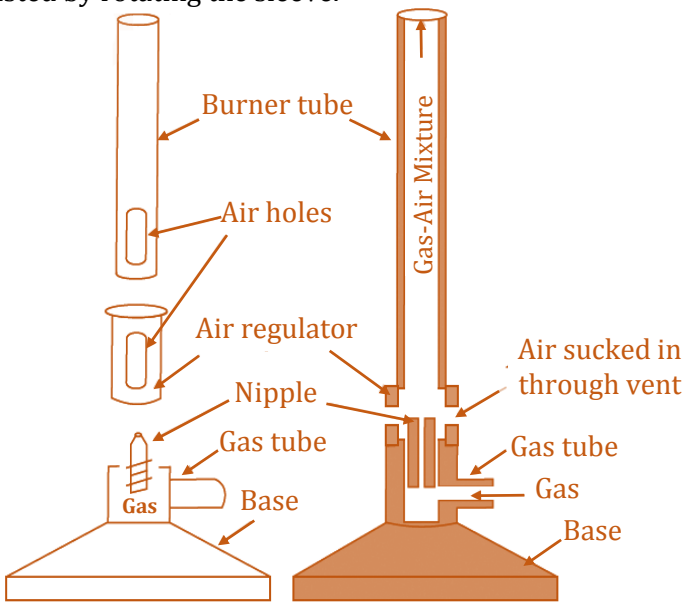


Fig. : Parts of Bunsen burner

If the air vent is closed and the gas is ignited, the flame will be large and luminous (smoky and yellow in colour). The light emitted by the flame is due to the radiations given off by the hot carbon particles of partially burnt fuel. The temperature of the flame in this situation is low. If adjustment of sleeve on vent is such that gas mixed with air is fed into the flame, the flame becomes less luminous and finally turns blue. When the flow of air is correctly adjusted, the temperature of the flame becomes quite high. This is called non-luminous flame. Various zones of flame are shown below in Fig.

Three distinctly visible parts of the Bunsen flame are described below :

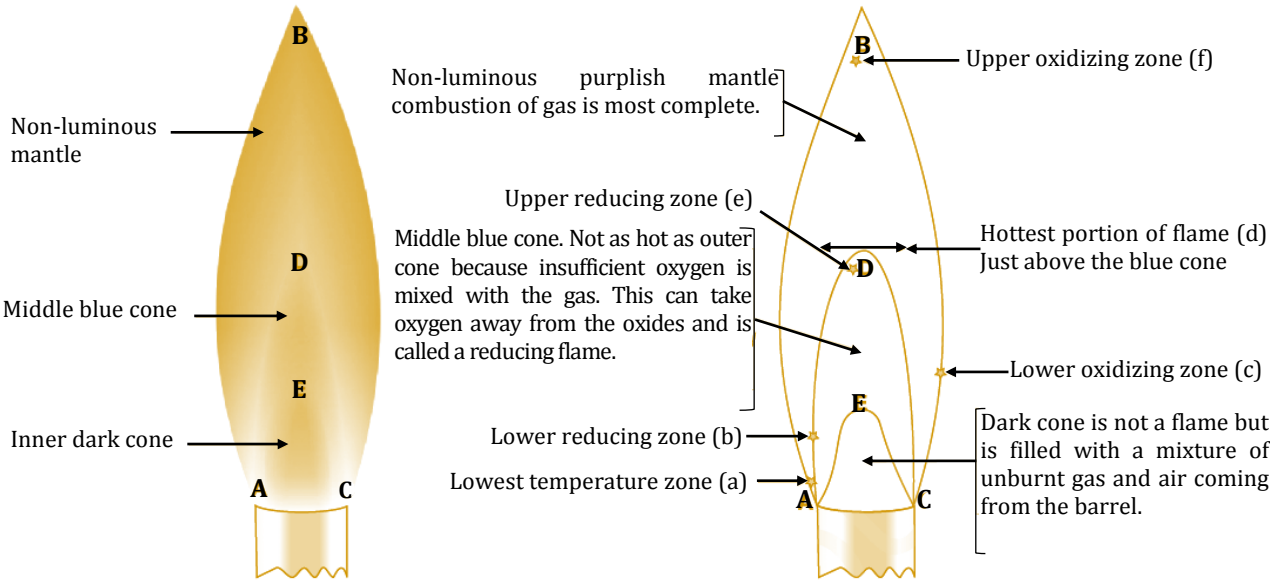


Fig. : Zones of flame of Bunsen burner

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(B) Principal parts of Bunsen flame

1. **The Inner Dark Cone, A E C**
   This is innermost dark cone, which is just above the burner tube. It consists of unburnt gases. This zone is the coldest zone of the flame and no combustion takes place here.
2. **The Middle Blue Cone, A D C E A**
   This is middle part of the flame. This becomes luminous when the air vent is slightly closed. Luminosity of this part is due to the presence of unburnt carbon particles produced by decomposition of some gas. These particles get heated up to incandescence and glow but do not burn. Since the combustion is not complete in this part, the temperature is not very high.
3. **The Outer Non-luminous Mantle, A B C D A**
   This is purplish outer cone. It is the hottest part of the flame. It is in direct contact with the atmosphere and combustion is quite complete in this zone.

Bunsen identified six different regions in these three principal parts of the flame:

(i) The upper oxidising zone (f)
Its location is in the non-luminous tip of the flame which is in the air. In comparison to inner portions of the flame large excess of oxygen is present here. The temperature is not as high as in region (c) described below. It may be used for all oxidation processes in which highest temperature of the flame is not required.

(ii) Upper reducing zone (e)
This zone is at the tip of the inner blue cone and is rich in incandescent carbon. It is especially useful for reducing oxide incrustations to the metals.

(iii) Hottest portion of flame (d)
It is the fusion zone. It lies at about one-third of the height of the flame and is approximately equidistant from inside and outside of the mantle i.e. the outermost cone of the flame. Fusibility of the substance can be tested in this region. It can also be employed for testing relative volatility of substances or a mixture of substances.

(iv) Lower oxidising zone (c)
It is located on the outer border of the mantle near the lower part of the flame and may be used for the oxidation of substances dissolved in beads of borax or sodium carbonate etc.

(v) Lower reducing zone (b)
It is situated in the inner edge of the outer mantle near to the blue cone and here reducing gases mix with the oxygen of the air. It is a less powerful reducing zone than (e) and may be employed for the reduction of fused borax and similar beads.

(vi) Lowest temperature zone (a)
Zone (a) of the flame has lowest temperature. It is used for testing volatile substances to determine whether they impart colour to the flame.

(C) Striking Back of The Bunsen Burner

Striking back is the phenomenon in which flame travels down the burner tube and begins to burn at the nozzle near the base. This happens when vents are fully open. The flow of much air and less gas makes the flame become irregular and it strikes back.

The tube becomes very hot and it may produce burns on touching. This may melt attached rubber tube also. If it happens, put off the burner and cool it under the tap and light it again by keeping the air vent partially opened.

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Dry tests:

Systematic Analysis Of Cations

The tests for cations may be carried out according to the following scheme.

Step - I : Preliminary Examination of the Salt for Identification of Cation

1. Colour Test

Observe the colour of the salt carefully, which may provide useful information about the cations. Table 12 gives the characteristic colours of the salts of some cations.

Table 12 Characteristic colours of the some metal ions

Ion	Confirmatory Test
Light green	$Fe^{2\oplus}$
Yellow is Brown	$Fe^{3\oplus}$
Blue	$Cu^{2\oplus}$
Bright green	$Ni^{2\oplus}$
Blue, Red Violet, Pink	$Co^{2\oplus}$
Light pink	$Mn^{2\oplus}$

2. Dry Heating Test

(i) Take about 0.1 g of the dry salt in a clean and dry test tube.

(ii) Heat the above test tube for about one minute and observe the colour of the residue when it is hot and also when it becomes cold. Observation of changes gives indications about the presence of cations, which may not be taken as conclusive evidence (see Table 13).

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Table 13 : Inferences from the colour of the salt in cold and on heating

Colour when cold	Colour when hot	Inference
Blue	White	$Cu^{2\oplus}$
Green	Dirty white or yellow	$Fe^{2\oplus}$
White	Yellow	$Zn^{2\oplus}$
Pink	Blue	$Co^{2\oplus}$

3. Flame Test

The chlorides of several metals impart characteristic colour to the flame because they are volatile in non-luminous flame. This test is performed with the help of a platinum wire as follows :

(i) Make a tiny loop at one end of a platinum wire.

(ii) To clean the loop dip it into concentrated hydrochloric acid and hold it in a non-luminous flame (Fig.).

(iii) Repeat step (ii) until the wire imparts no colour to the flame.

(iv) Put 2-3 drops of concentrated hydrochloric acid on a clean watch glass and make a paste of a small quantity of the salt in it.

(v) Dip the clean loop of the platinum wire in this paste and introduce the loop in the non-luminous (oxidising) flame (Fig.).

(vi) Observe the colour of the flame first with the naked eye and then through a blue glass and identify the metal ion with the help of Table 14.

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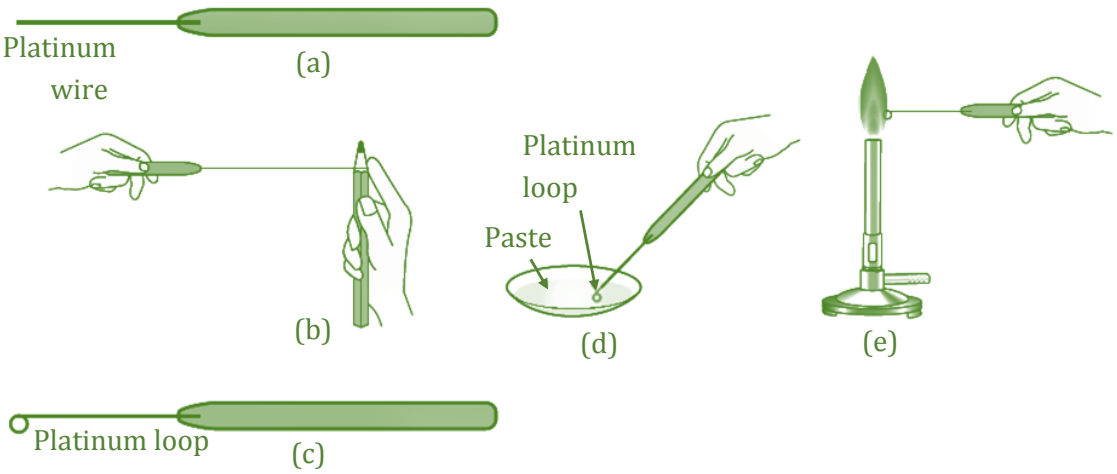


Fig. : Performing flame test

Table 14 : Inference from the flame test

Colour of the flame observed by naked eye	Colour of the flame observed through blue glass	Inference
Green flame with blue centre	Same colour as observed without glass	$Cu^{2\oplus}$
Crimson red	Purple	$Sr^{2\oplus}$
Apple green	Bluish green	$Ba^{2\oplus}$
Brick red	Green	$Ca^{2\oplus}$

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4. Borax Bead Test

This test is employed only for coloured salts because borax reacts with metal salts to form metal borates or metals, which have characteristic colours.

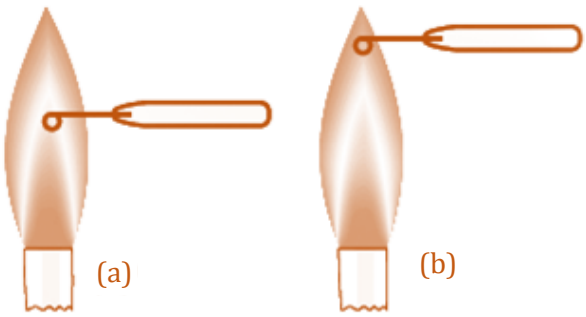


Fig. : Borax bead test
(a) Heating in reducing flame
(b) Heating in oxidizing flame

(i) To perform this test make a loop at the end of the platinum wire and heat it in a flame till it is red hot.

(ii) Dip the hot loop into borax powder and heat it again until borax forms a colourless transparent bead on the loop. Before dipping the borax bead in the test salt or mixture, confirm that the bead is transparent and colourless. If it is coloured this means that, the platinum wire is not clean. Then make a fresh bead after cleaning the wire.

(iii) Dip the bead in a small quantity of the dry salt and again hold it in the flame.

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(iv) Observe the colour imparted to the bead in the non - luminous flame as well as in the luminous flame while it is hot and when it is cold (Fig.).

(v) To remove the bead from the platinum wire, heat it to redness and tap the platinum wire with your finger. (Fig.).

On heating, borax loses its water of crystallisation and decomposes to give sodium metaborate and boric anhydride.

$$\underset{\text{Borax}}{Na_2B_4O_7.10H_2O} \longrightarrow Na_2B_4O_7 + 10H_2O$$

$$Na_2B_4O_7 \longrightarrow \underset{\text{Sodium metaborate}}{2NaBO_2} + \underset{\text{Boric anhydride}}{B_2O_3}$$

On treatment with metal salt, boric anhydride forms metaborate of the metal which gives different colours in oxidising and reducing flame. For example, in the case of copper sulphate, following reactions occur.

$$CuSO_4 + B_2O_3 \xrightarrow{\text{Non-luminous flame}} \underset{\substack{\text{Cupric metaborate}\\ \text{Blue-green}}}{Cu(BO_2)_2} + SO_3$$

Two reactions may take place in the reducing flame:

(i) The blue $Cu(BO_2)_2$ is reduced to colourless cuprous metaborate as follows:

$$2Cu(BO_2)_2 + 2NaBO_2 + C \xrightarrow{\text{Luminous flame}} 2CuBO_2 + Na_2B_4O_7 + CO$$

or

(ii) Cupric metaborate may be reduced to metallic copper and the bead appears red and opaque.

$$2Cu(BO_2)_2 + 4NaBO_2 + 2C \xrightarrow{\text{Luminous flame}} 2Cu + 2Na_2B_4O_7 + 2CO$$

The preliminary identification of metal ion can be made from Table 15.

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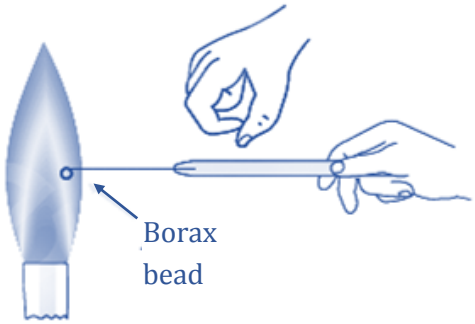


Fig. : Removing borax bead

Table 15 : Inference from the borax bead test

Heating in oxidizing (non-luminous) flame		Heating in reducing (luminous) flame		Inference
Colour of the salt bead		Colour of the salt bead		
In cold	In hot	In cold	In hot	
Blue	Green	Red opaque	Colourless	$Cu^{2\oplus}$
Reddish brown	Violet	Grey	Grey	$Ni^{2\oplus}$
Light violet	Light violet	Colourless	Colourless	$Mn^{2\oplus}$
Yellow	Yellowish brown	Green	Green	$Fe^{3\oplus}$

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5. Charcoal Cavity Test

Metallic carbonate when heated in a charcoal cavity decomposes to give corresponding oxide. The oxide appears as a coloured residue in the cavity. Sometimes oxide may be reduced to metal by the carbon of the charcoal cavity.

The test may be performed as follows:

(i) Make a small cavity in a charcoal block with the help of a charcoal borer [Fig. (a)].

(ii) Fill the cavity with about 0.2 g of the salt and about 0.5 g of anhydrous sodium carbonate.

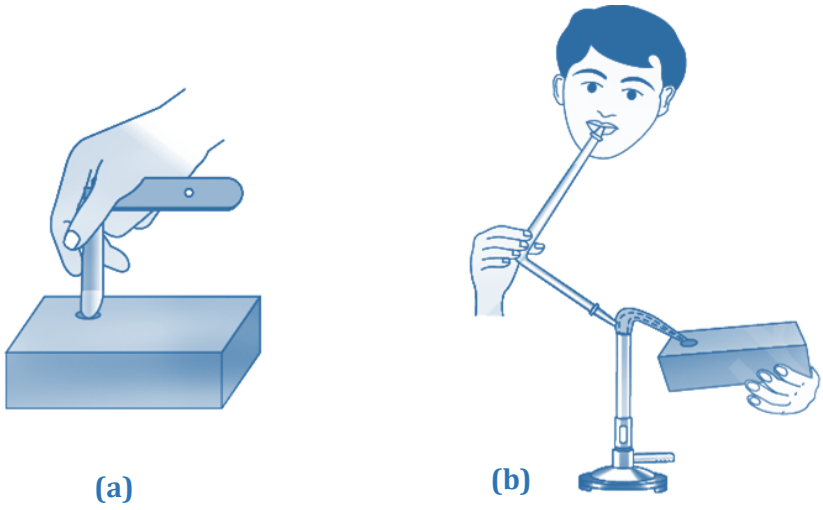


Fig. : (a) Making charcoal cavity (b) Heating salt in the cavity

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(iii) Moisten the salt in the cavity with one or two drops of water, otherwise salt/mixture will blow away.

(iv) Use a blowpipe to heat the salt in a luminous (reducing) flame and observe the colour of oxide/ metallic bead formed in the cavity both when hot and cold [Fig. (b)]. Obtain oxidising and reducing flame as shown in Fig. a and b.

(v) Always bore a fresh cavity for testing the new salt.

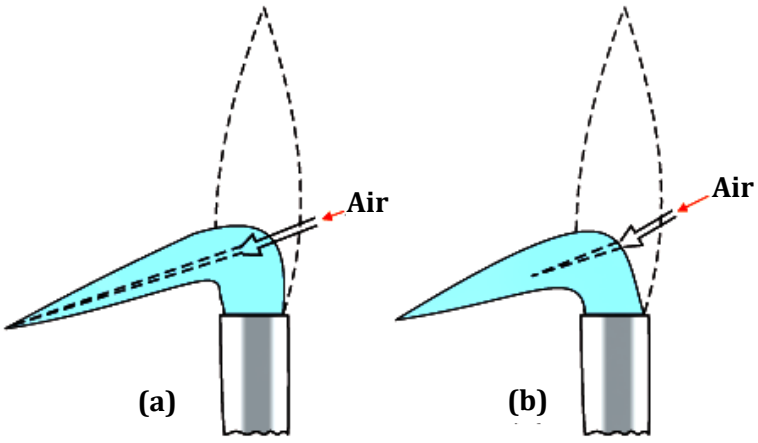


Fig. : Obtaining oxidizing and reducing flame
(a) Oxidizing flame; and (b) Reducing flame

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When test is performed with $CuSO_4$, the following change occurs.

$$CuSO_4 + Na_2CO_3 \xrightarrow{Heat} CuCO_3 + Na_2SO_4$$

$$CuCO_3 \xrightarrow{Heat} CuO + CO_2$$

$$CuO + C \xrightarrow{Heat} Cu + CO$$

Red colour

In case of $ZnSO_4$:

$$ZnSO_4 + Na_2CO_3 \xrightarrow{Heat} ZnCO_3 + Na_2SO_4$$

$$ZnCO_3 \xrightarrow{Heat} ZnO + CO_2$$

Yellow when hot,
White when cold

The metal ion can be inferred from Table 16.

Table 16 : Inference from the charcoal cavity test

Observations	Inference
Yellow residue when hot and grey metal when cold	$Pb^{2\oplus}$
White residue with the odour of garlic	$As^{3\oplus}$
Brown residue	$Cd^{2\oplus}$
Yellow residue when hot and white when cold	$Zn^{2\oplus}$

6. Cobalt Nitrate Test

If the residue in the charcoal cavity is white, cobalt nitrate test is performed.

(i) Treat the residue with two or three drops of cobalt nitrate solution.

(ii) Heat it strongly in non-luminous flame with the help of a blow pipe and observe the colour of the residue.

On heating, cobalt nitrate decomposes into cobalt (II) oxide, which gives a characteristic colour with metal oxide present in the cavity.

Thus, with ZnO, Al_2O_3 and MgO, the following reactions occur.

$$2Co(NO_3)_2 \xrightarrow{Heat} 2CoO + 4NO_2 + O_2$$

$$CoO + ZnO \longrightarrow CoO.ZnO$$

Green

$$CoO + MgO \longrightarrow CoO.MgO$$

Pink

$$CoO + Al_2O_3 \longrightarrow CoO.Al_2O_3$$

Blue

Step-II : Wet Tests for Identification of Cations

The cations indicated by the preliminary tests given above are confirmed by systematic analysis given below. The first essential step is to prepare a clear and transparent solution of the salt. This is called original solution. It is prepared as follows:

Preparation of Original Solution (O.S.)

To prepare the original solution, following steps are followed one after the other in a systematic order. In case the salt does not dissolve in a particular solvent even on heating, try the next solvent.

The following solvents are tried:

1. Take a little amount of the salt in a clean boiling tube and add a few mL of distilled water and shake it. If the salt does not dissolve, heat the content of the boiling tube till the salt completely dissolves.

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2. If the salt is insoluble in water as detailed above, take fresh salt in a clean boiling tube and add a few mL of dil. HCl to it. If the salt is insoluble in cold, heat the boiling tube till the salt is completely dissolved.
3. If the salt does not dissolve either in water or in dilute HCl even on heating, try to dissolve it in a few mL of conc. HCl by heating.
4. If salt does not dissolve in conc. HCl, then dissolve it in dilute nitric acid.
5. If salt does not dissolve even in nitric acid then a mixture of conc. HCl and conc. HNO_3 in the ratio 3 : 1 is tried. This mixture is called aqua regia. A salt not soluble in aqua regia is considered to be an insoluble salt.

Heating Effect:

1. Heating effect of carbonate & bicarbonate salts:

(a) Heating effect of carbonate salts:

Metal carbonate $\xrightarrow{\Delta}$ metal oxide + $CO_2\uparrow$

(i) $MCO_3 \xrightarrow{\Delta} MO + CO_2$ [M = Be, Mg, Ca, Sr, Ba]

(ii) $MgCO_3 \xrightarrow{\Delta} MgO + CO_2$

(iii) $\underbrace{Cu(OH)_2.CuCO_3}_{\text{Basic Cu(II) carbonate}} \xrightarrow{\Delta} \underset{\text{(black)}}{2CuO} + CO_2 + H_2O \xrightarrow{\text{High temperature}} \underset{\text{(red)}}{Cu_2O} + O_2$

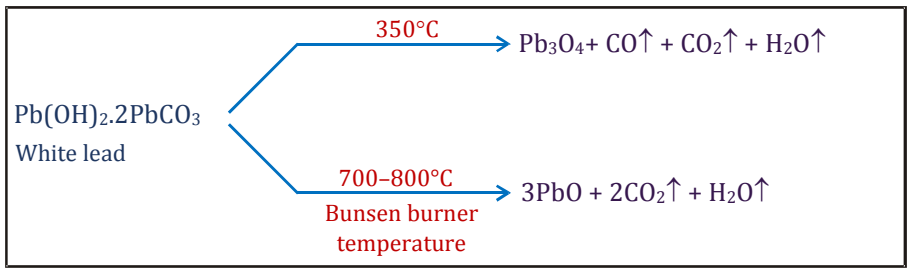
(iv) $\underset{\text{white}}{ZnCO_3} \xrightarrow{\Delta} \underset{\substack{\text{Yellow(hot)}\\ \text{white(cold)}}}{ZnO} + CO_2$

(v) $PbCO_3 \xrightarrow{\Delta} \underset{\text{Yellow}}{PbO} + CO_2$

(vi) $Li_2CO_3 \xrightarrow{\Delta} Li_2O + CO_2$

(vii) $\boxed{(NH_4)_2CO_3 \xrightarrow{\Delta} 2NH_3 + H_2O + CO_2 \quad \text{Very Important}}$

(viii)



(ix) All carbonates **except** (Na, K, Rb, Cs) decompose on heating giving CO_2

(x) Carbonates salts of (Na, K, Rb, Cs) do not decompose on heating, they are melt on high temperature.

(xi) Oxides of heavier metals are less stable so further decompose into metal & oxygen

(xii) $\underset{\substack{\text{white}\\ \text{(yellowishwhite)}}}{Ag_2CO_3} \xrightarrow{\Delta} \underset{\text{Black}}{2Ag} + CO_2 + \frac{1}{2}O_2$

(xi) $HgCO_3 \xrightarrow{\Delta} Hg + \frac{1}{2}O_2\uparrow + CO_2$

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(b) Heating effect of bicarbonate:

(i) Metal bicarbonate $\xrightarrow{\Delta}$ metal carbonate + $CO_2\uparrow$ $\longrightarrow$ metal oxide + CO_2
(except (Na, K, Rb, Cs))

(ii) [General reaction $2HCO_3^{\ominus} \xrightarrow{\Delta} CO_3^{2\ominus} + H_2O + CO_2$]

(iii) All bicarbonates decompose to give carbonates and CO_2. eg.

(iv) $2NaHCO_3 \xrightarrow{\Delta} Na_2CO_3 + CO_2 + H_2O$

(v) $Mg(HCO_3)_2 \xrightarrow{\Delta} MgO + 2CO_2 + H_2O$

2. Heating effect of hydrated Sulphate salts:

(i) Metal Sulphate $\xrightarrow{\Delta}$ Metal Oxide + SO_3 $\xrightarrow{T>800^\circ C}$ $SO_2 + \frac{1}{2}O_2 \xrightarrow{1100^\circ C} \frac{1}{2}Cu_2O + \frac{1}{2}O_2$
(M=Be,Zn,Mg,Ca,Cu,Pb)

(ii) $CuSO_4.5H_2O \xrightarrow[-4H_2O]{100^\circ C}$ $CuSO_4.H_2O$ $\xrightarrow[-H_2O]{220^\circ C}$ $CuSO_4$ $\xrightarrow{>800^\circ C}$ $CuO + SO_2 + \frac{1}{2}O_2$
Blue vitriol — Bluish white — White

$CuSO_4 \xrightarrow{750^\circ C} CuO + SO_3$ (Black)

(iii) $FeSO_4.7H_2O \xrightarrow[-7H_2O]{300^\circ C} FeSO_4 \xrightarrow[\text{Strongly}]{\Delta} Fe_2O_3 + SO_2 + SO_3$ (very important)
Green vitriol

(iv) $Fe_2(SO_4)_3 \xrightarrow{>800^\circ C} Fe_2O_3 + 3SO_3$

(v) $CaSO_4.2H_2O \xrightarrow{120^\circ C\text{-}150^\circ C} (CaSO_4.\frac{1}{2}H_2O) + 1\frac{1}{2}H_2O$
gypsum — (Plaster of paris) or calcium sulphate hemihydrate

$CaSO_4.2H_2O \xleftarrow{1\frac{1}{2}H_2O}$ (Plaster of paris) $\xrightarrow{\Delta} CaSO_4 + \frac{1}{2}H_2O$

$CaSO_4.2H_2O$: another crystallographic form of gypsum hard & porous

$CaSO_4 + \frac{1}{2}H_2O$: Dead burnt $\xrightarrow{1100^\circ C} CaO + SO_3$

(vi) $ZnSO_4.7H_2O \xrightarrow[-H_2O]{70^\circ C} ZnSO_4.6H_2O \xrightarrow[-5H_2O]{70^\circ\text{-}200^\circ C} ZnSO_4.H_2O \xrightarrow[-H_2O]{>200^\circ C} ZnSO_4 \xrightarrow{\Delta>800^\circ C} ZnO + SO_2\,\frac{1}{2}O_2$

(vii) $MgSO_4.7H_2O \xrightarrow[-7H_2O]{\Delta} MgSO_4\downarrow$ [Same as $ZnSO_4$]
epsom salt

(viii) $2NaHSO_3 \xrightarrow{>800^\circ C} Na_2S_2O_5 + \xrightarrow{\Delta} Na_2SO_3 + SO_2$
$Na_2S_2O_5 + \longrightarrow Na_2S + Na_2SO_4$

(ix) $2NaHSO_4 \xrightarrow{>800^\circ C} Na_2SO_4 + H_2O + SO_3$

(x) $Na_2S_2O_3.5H_2O \xrightarrow{220^\circ C} Na_2S_2O_3 + 5H_2O$
$Na_2S_2O_3 \xrightarrow[\text{(From 4 moles)}]{\Delta} 3Na_2SO_4$

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3. Heating effect of nitrate salts:

(i) Metal nitrate $\xrightarrow{>800^\circ C}$ metal oxide + NO_2 + O_2

(ii) $2M(NO_3)_2 \xrightarrow{>800^\circ C} 2MO + 4NO_2 + O_2$

[M = all bivalent metals eg. $Zn^{2\oplus}$, $Mg^{2\oplus}$, $Sr^{2\oplus}$, $Ca^{2\oplus}$, $Ba^{2\oplus}$, $Cu^{2\oplus}$, $Pb^{2\oplus}$]

(iii) $2LiNO_3 \xrightarrow{>800^\circ C} Li_2O + 2NO_2 + \frac{1}{2}O_2$

(iv) $MNO_3 \xrightarrow[\text{[M=Na,K,Rb,Cs]}]{>800^\circ C} MNO_2 + \frac{1}{2}O_2 \xrightarrow{\text{high temperature}} M_2O + N_2 + \frac{3}{2}O_2$

(v) $NaNO_3 \xrightarrow{500^\circ C} NaNO_2 + \frac{1}{2}O_2$

(vi) $NaNO_2 \xrightarrow{800^\circ C} Na_2O + N_2 + \frac{3}{2}O_2$

(vii) $BeCl_2 \xrightarrow{N_2O_4} Be(NO_3)_2.2N_2O_4 \xrightarrow[\text{under vaccum}]{\text{Warm to 50°C}} Be(NO_3)_2 \xrightarrow{125^\circ C} [Be_4O(NO_3)_6]$ (Basic beryllium nitrate)

$Be(NO_3)_2 \xrightarrow{700-800^\circ C} BeO + 2NO_2 + \frac{1}{2}O_2$

Exception : If formed oxide is of heavier metal then it being less stable and further decomposed into metal and oxygen.

(viii) $Hg(NO_3)_2 \xrightarrow{>800^\circ C} Hg + 2NO_2 + O_2$

(ix) $2AgNO_3 \xrightarrow{>800^\circ C} 2Ag + 2NO_2 + O_2$

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4. Heating effect of ammonium salts:

If anionic part is oxdising in nature, then N_2 will be the product (some times N_2O).

(i) $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} N_2 + Cr_2O_3 + 4H_2O$ {This reaction is used for making artificial volcano}
(orange solid) (green)

(ii) $NH_4NO_2 \xrightarrow{\Delta} N_2 + 2H_2O$

(iii) $NH_4NO_3 \xrightarrow{\Delta} N_2O + 2H_2O$

(iv) $2NH_4ClO_4 \xrightarrow{\Delta} N_2 + Cl_2 + 2O_2 + 4H_2O$

(v) $2NH_4IO_3 \xrightarrow{\Delta} N_2 + I_2 + O_2 + 4H_2O$

If anionic part weakly oxidising or non oxidising in nature then NH_3 will be the product.

(i) $(NH_4)_2HPO_4 \xrightarrow{\Delta} HPO_3 + H_2O + 2NH_3$

(ii) $(NH_4)_2SO_4 \xrightarrow{\Delta} NH_3 + H_2SO_4$

(iii) $2(NH_4)_3PO_4 \xrightarrow{\Delta} 2NH_3 + P_2O_5 + 3H_2O$

(iv) $(NH_4)_2CO_3 \xrightarrow{\Delta} 2NH_3 + H_2O + CO_2$

(v) $NH_4Cl \xrightarrow{\Delta} NH_3\uparrow + HCl\uparrow$

(vi) $(NH_4)_2S \xrightarrow{\Delta} NH_3\uparrow + H_2S\uparrow$

(vii) $NH_4F \xrightarrow{\Delta} NH_3 + HF$

(viii) $(NH_4)_2MoO_4 \xrightarrow{\Delta} NH_3\uparrow + MoO_3 + H_2O$

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5. Heating effect of phosphate salts:

(i) 1° Phosphate salts gives metaphosphate salt on heating.

$NaH_2PO_4 \xrightarrow{\Delta} H_2O + NaPO_3$

(ii) 2° Phosphate salts gives pyrophosphate

$Na_2HPO_4 \xrightarrow{\Delta} H_2O + Na_4P_2O_7$

(iii) 3° Phosphate salt have no heating effect

$Na_3PO_4 \xrightarrow{\Delta}$ No effect

(iv) $\underset{\text{microcosmic salt}}{Na(NH_4)HPO_4.4H_2O} \xrightarrow[-4H_2O]{\Delta} NaNH_4HPO_4 \xrightarrow{\text{High temp.}} NaPO_3 + NH_3 + H_2O$

(v) $2Mg(NH_4)PO_4 \xrightarrow{\Delta} Mg_2P_2O_7 + 2NH_3 + H_2O$

6. Heating effect of Halides salts:

(i) $2FeCl_3 \xrightarrow{\Delta} 2FeCl_2 + Cl_2$

(ii) $AuCl_3 \xrightarrow{\Delta} AuCl + Cl_2$

(iii) $Hg_2Cl_2 \xrightarrow{\Delta} HgCl_2 + Hg$

(iv) $NH_4Cl \xrightarrow{\Delta} NH_3 + HCl$

(v) $Pb(SCN)_4 \xrightarrow{\Delta} Pb(SCN)_2 + (SCN)_2$

(vi) $PbCl_4 \xrightarrow{\Delta} PbCl_2 + Cl_2$

(vii) $PbBr_4 \xrightarrow{\Delta} PbBr_2 + Br_2$ **[PbI_4 does not exists]**

(viii) $\underset{\text{scarlet red}}{HgI_2} \underset{\text{On Rubbing}}{\overset{127^\circ C}{\rightleftharpoons}} \underset{\text{yellow}}{HgI_2}$

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7. Heating effect of hydrated chloride salts:

(i) $MgCl_2.6H_2O \xrightarrow{\Delta} MgO + 2HCl + 5H_2O$

(ii) $2FeCl_3.6H_2O \xrightarrow{\Delta} Fe_2O_3 + 6HCl + 9H_2O$

(iii) $2AlCl_3.6H_2O \xrightarrow{\Delta} Al_2O_3 + 6HCl + 9H_2O$

(iv) $\underset{\text{Pink}}{CoCl_2.6H_2O} \xrightarrow[-2H_2O]{50^\circ C} \underset{\text{blue}}{CoCl_2.4H_2O} \xrightarrow[-2H_2O]{58^\circ C} \underset{\text{Red violet}}{CoCl_2.2H_2O} \xrightarrow[-2H_2O]{140^\circ C} \underset{\text{Blue}}{CoCl_2}$

Hydrated $Co^{2\oplus}$ salt - Pink Anhydrous $Co^{2\oplus}$ salt - Blue

8. Heating effect of Oxide:

(i) $2Ag_2O \xrightarrow[300^\circ C]{\Delta} 4Ag + O_2$

(ii) $\underset{\text{white}}{ZnO} \underset{\text{Cold}}{\overset{\text{Hot}}{\rightleftharpoons}} \underset{\text{yellow}}{ZnO}$

(iii) $PbO_2 \xrightarrow{\Delta} PbO + \frac{1}{2}O_2$

(iv) $\underset{\text{yellow}}{PbO(\text{Massicot})} \xrightarrow[\text{Cooled \& powdered}]{\text{Fused}} \underset{\text{Litharge (red)}}{PbO}$

(v) $3MnO_2 \xrightarrow{900^\circ C} Mn_3O_4 + O_2$

(vi) $\underset{\text{Red lead}}{Pb_3O_4} \underset{350^\circ C}{\overset{500^\circ C}{\rightleftharpoons}} \underset{\text{Litharge}}{6PbO} + O_2$

(vii) $2CrO_5 \xrightarrow{\Delta} Cr_2O_3 \frac{7}{2} + O_2$

(viii) $\underset{\text{white}}{K_2O} \underset{\text{Cold}}{\overset{\Delta}{\rightleftharpoons}} \underset{\text{yellow}}{K_2O}$

(ix) $I_2O_5 \xrightarrow{\Delta} I_2 + \frac{5}{2}O_2$

(x) $\underset{\text{yellow}}{HgO} \xrightarrow{\Delta} \underset{\text{red}}{HgO} \xrightarrow[\text{strong heating}]{400^\circ C} Hg + \frac{1}{2}O_2$

(xi) $2CrO_3 \xrightarrow{420^\circ C} Cr_2O_3 + \frac{3}{2}O_2$

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9. Heating effect of permanganate:

$$\underset{\text{dark purple (almost black)}}{2KMnO_4} \xrightarrow{513k} \underset{\text{(green)}}{K_2MnO_4} + \underset{\text{(black)}}{MnO_2} + O_2$$

10. Heating effect of dichromate & chromate salts:

$$\underset{\text{orange}}{2K_2Cr_2O_7} \xrightarrow{\Delta} \underset{\text{yellow}}{2K_2CrO_4} + \underset{\text{green}}{Cr_2O_3} + \frac{3}{2}O_2\uparrow$$

11. Heating effect of Acids:

(i) $\underset{\text{conc.}}{2HNO_3} \xrightarrow{\Delta} H_2O + 2NO_2 + \frac{1}{2}O_2$

(ii) $H_2SO_4 \xrightarrow{444^\circ C} H_2O + SO_3$

(iii) $H_2SO_4 \xrightarrow{>800^\circ C} H_2O + SO_2 + \frac{1}{2}O_2$

(iv) $3H_2SO_3 \longrightarrow 2H_2SO_4 + S\downarrow + H_2O$

(v) $3HNO_2 \longrightarrow HNO_3 + 2NO + H_2O$

(vi) $HClO_3 \longrightarrow HClO_4 + ClO_2 + H_2O$

(vii) $3HOCl \longrightarrow 2HCl + HClO_3$

(viii) $4H_3PO_3 \xrightarrow[200^\circ C]{\Delta} 3H_3PO_4 + PH_3$

(ix) $2H_3PO_2 \longrightarrow H_3PO_4 + PH_3$

(x) $2NaH_2PO_2 \longrightarrow Na_2HPO_4 + PH_3$

(xi) $H_2C_2O_4 \xrightarrow{\Delta} H_2O + CO + CO_2$

} Undergoes disproportionation reaction

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(xii) $H_3PO_4 \xrightarrow{220^\circ C} H_4P_2O_7 \xrightarrow{320^\circ C} 4HPO_3 \xrightarrow{>320^\circ C} 2P_2O_5 + 2H_2O$

(xiii) $H_3BO_3 \xrightarrow{100^\circ C} 4HBO_2 \xrightarrow{140^\circ C} H_2B_4O_7 \xrightarrow[\text{hot}]{\text{Red}} 2B_2O_3 + H_2O$

12. Heating effects of acetate salts:

$$\text{Metal acetate} \xrightarrow{\Delta} \underset{\text{(M = LiBe Mg)}}{\text{Metal Carbonate}} + CH_3COCH_3$$

$$\downarrow \Delta$$

$$\text{Metal oxide} + CO_2$$

(i) $Pb(OAc)_2 \xrightarrow{\Delta} PbO + CO_2 + CH_3COCH_3$

(ii) $Mg(OAc)_2 \xrightarrow{\Delta} MgO + CO_2 + CH_3COCH_3$

(iii) $Be(OAc)_2 \xrightarrow{\Delta} BeO + CO_2 + CH_3COCH_3$

(iv) $Ca(OAc)_2 \xrightarrow{\Delta} CaCO_3 + CH_3COCH_3$

(v) $Ba(OAc)_2 \xrightarrow{\Delta} BaCO_3 + CH_3COCH_3$

(vi) $CH_3CO_2K \xrightarrow{\Delta} K_2CO_3 + CH_3COCH_3$

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13. Heating effects of oxalate salts:

$$\text{Metal oxalate} \xrightarrow{\Delta} \underset{(M = LiBe\ Mg)}{\text{Metal Carbonate}} + CO + CO_2$$

$$\downarrow \Delta$$

$$\text{Metal oxide} + CO_2$$

(i) $7Na_2C_2O_4 \xrightarrow{\Delta} 7Na_2CO_3 + 2CO_2 + 3CO + 2C$

(ii) $SnC_2O_4 \xrightarrow{\Delta} SnO + CO_2 + CO$

(iii) $FeC_2O_4 \xrightarrow{\Delta} FeO + CO + CO_2$

(iv) $Ag_2C_2O_4 \xrightarrow{\Delta} 2Ag + 2CO_2$

(v) $HgC_2O_4 \xrightarrow{\Delta} Hg + 2CO_2$

14. Heating effects of formate salts:

(i) $HCO_2Na \xrightarrow{350^\circ C} Na_2C_2O_4 + H_2\uparrow$

(ii) $HCOOAg \xrightarrow{\Delta} HCOOH + 2Ag + \underbrace{\frac{1}{2}O_2 + CO}_{\downarrow}$

CO_2

(iii) $(HCOO)_2Hg \longrightarrow HCOOH + Hg + \underbrace{\frac{1}{2}O_2 + CO}_{\downarrow}$

CO_2

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